



सत्यमेव जयते

भारत सरकार

Government of India

विद्युत् मंत्रालय

Ministry of Power

पूर्वी क्षेत्रीय विद्युत् समिति

Eastern Regional Power Committee

सं/NO. ERPC/EE/OPERATION/2024/1787

दिनांक/DATE: 26.03.2024

सेवा में /To

संलग्न सूची के अनुसार /As per list enclosed.

विषय : 14.03.2024 (गुरुवार) को ईआरपीसी सचिवालय, कोलकाता में भौतिक रूप से आयोजित 213वीं ओसीसी बैठक का कार्यवृत्त - संबंध में।

Sub: Minutes of 213th OCC Meeting held on 14.03.2024 (Thursday) physically at ERPC Secretariat, Kolkata - reg.

महोदय/महोदया,

Sir(s)/Madam,

कृपया अपनी जानकारी और आवश्यक कार्रवाई के लिए 14.03.2024 (गुरुवार) को ईआरपीसी सचिवालय, कोलकाता में भौतिक रूप से 10:30 बजे आयोजित 213वीं ओसीसी बैठक के संलग्न कार्यवृत्त देखें। यह ईआरपीसी वेबसाइट (www.erpc.gov.in) पर भी उपलब्ध है।

Please find enclosed **Minutes of 213th OCC Meeting** held on 14.03.2024 (Thursday) physically at ERPC Secretariat, Kolkata at 10:30 hrs for your kind information and necessary action. The same is also available at ERPC website (www.erpc.gov.in).

टिप्पणियाँ, यदि कोई हों, कृपया यथाशीघ्र इस कार्यालय को अग्रेषित करें।

Observations, if any, may please be forwarded to this office at the earliest.

भवदीय /Yours faithfully,

नि.स. मंडल
26.3.2024
(N.S. Mondal)

Member Secretary/सदस्य सचिव)

LIST OF ADDRESSES:

1. CHIEF ENGINEER (TRANS., O&M), BSPTCL, PATNA, (FAX NO. 0612-2504557/2504937)
2. CHIEF ENGINEER (System Operation), BSPTCL, PATNA, (FAX NO. 0612-2504557/2504937)
3. CHIEF ENGINEER, TRANSMISSION (O&M), JUSNL, RANCHI (FAX NO.-0651-2490486/2490863)
4. CHIEF ENGINEER, TVNL, DORANDA, RANCHI - 834102 (FAX NO. 06544-225414)
5. CHIEF LOAD DISPATCHER, SLDC, OPTCL, BHUBANESWAR (FAX NO.0674-2748509)
6. CHIEF GENERAL MANAGER (O&M), OPTCL, BHUBANESWAR
7. SR. GENERAL MANAGER (PP), GRIDCO, JANPATH, BHUBANESWAR (0674-2547180)
8. DIRECTOR (OPERATION), IB TPS, AT/PO BANHARPALI, JHARSUGUDA, (FAX NO. 06645-222225/222230)
9. GENERAL MANAGER, TTPS, TALCHER, (FAX NO. 06760-243212)
10. SR. GENERAL MANAGER (ELECTRICAL), OHPC LTD., BHUBANESWAR, (FAX NO.0674-2542102)
11. CHIEF ENGINEER, CLD, WBSETCL, HOWRAH, (FAX NO. 033-26886232)
12. CHIEF ENGINEER, CENTRAL PLANNING WING, WBSETCL, SALT LAKE (FAX NO.: 033-23591955)
13. CHIEF ENGINEER (PTR), WBSEDCL, SALT LAKE, KOLKATA (FAX:033-23345862)
14. CHIEF GENERAL MANAGER (OS), WBPDC, KOLKATA-98 (FAX NO. 033-23393286/2335-0516)
15. GM, KOLAGHAT TPS, WBPDC, KOLAGHAT (FAX NO.03228231280)
16. DGM (OPERATION), DPL, DURGAPUR, (FAX NO. 0343-2555052)
17. GM (SYS OPERATION), CESC, CHOWRINGHEE SQUARE, KOLKATA (FAX NO.033-22253756/22129871)
18. CHIEF ENGINEER, SLDC, DVC, HOWRAH (FAX NO. 033-2688-5094)
19. ADDL.CHIEF ENGINEER, SLDC, POWER DEPT., GOVT. OF SIKKIM, GANGTOK, (FAX NO. 03592-228186/201148/202284)
20. EXECUTIVE DIRECTOR, ERLDC, POSOCO, KOLKATA, (FAX NO. 033-2423-5809)
21. GENERAL MANAGER, FSTPP, NTPC, FARAKKA, (FAX NO. 03512-224214/226085/226124)
22. GENERAL MANAGER , KhSTPP, NTPC, KAHALGAON (FAX NO.06429-226082)
23. GENERAL MANAGER, TSTPP, NTPC, TALCHER, (FAX NO. 06760-249053)
24. GENERAL MANAGER (OS), POWERGRID, ER-II, KOLKATA(Fax no: 033-23572827)
25. GENERAL MANAGER , POWERGRID, ER-I, PATNA, (FAX NO.0612-2531192)
26. GENERAL MANAGER (O&M), POWERGRID, ODISHA PROJECTS, SAHID NAGAR, BHUBANESWAR – 751 007
27. MANAGING DIRECTOR, DRUK GREEN POWER CORPORATION, P.O. BOX -1351, THIMPU, BHUTAN —(FAX NO 00975- 2336411)
28. MANAGING DIRECTOR, BHUTAN POWER CORPORATION, P.O.BOX-580, THIMPU, BHUTAN (FAX NO. 00975-2333578)
29. CHIEF ENGINEER (O&M), TALA H.E.PROJECT, BHUTAN (FAX NO. 009752/324803)
30. EXECUTIVE DIRECTOR (O&M), NHPC, FARIDABAD (FAX No.:0129-2272413)

31. GENERAL MANAGER, TEESTA –V POWER STATION, NHPC, SINGTAM, EAST SIKKIM (FAX 03592 - 247377)
32. CHIEF ENGINEER, RANGIT POWER STATION, NHPC, P.O. RANGIT NAGAR, SOUTH SIKKIM (FAX NO.03595-259268)
33. SENIOR VICE PRESIDENT, PTC LTD., NBCC TOWERS, 15-BHIKAJI KAMA PLACE, NEW DELHI- 110066 (FAX NO. 011-41659504)
34. PLANT HEAD, ADHUNIK POWER & NATUARAL RESOURCES, JHARKHAND(FAX NO.: 0657-6628440)
35. AGM (OPERATION), MAITHON POWER LTD, DHANBAD (FAX: 08860004758)
36. VICE PRESIDENT(POWER), VEDANTA LIMITED, BHUBANESWAR- 751023 (FAX NO 0674-2302920)
37. CHIEF ELECTRICAL ENGINEER, EASTERN RAILWAY, KOLKATA-700 001 (FAX NO.: 033-22300446)
38. CHIEF ELECTRICAL ENGINEER, SOUTH EASTERN RAILWAY, KOLKATA-43 (FAX: 033-24391566)
39. DEPUTY DIRECTOR, EASTERN RPSO, SALT LAKE, KOLKATA- (FAX NO:033-23217075)
40. GENERAL MANAGER (O&M), NHPC LTD, FARIDABAD, FAX: 0129-2272413
41. ASSOCIATE VICE PRESIDENT, GMR KEL, BHUBANESWAR-751007. (FAX NO: 0674-2572794)
42. GM (SO & COMML), NTPC VVNL, NEW DELHI-110033. Fax:011-24367021
43. SHRI D. P. BHAGAVA, CHIEF CONSULTANT (O&M), TEESTA URJA LIMITED, NEW DELHI-110 001 (FAX:011-46529744)
44. SHRI BRAJESH KUMAR PANDE, PLANT HEAD, JITPL. (FAX:011-26139256-65)
45. DIRECTOR (NPC), CEA, NRPC BUILDING, KATWARIA SARAI, NEW DELHI- 110016
46. DGM (OS), HALDIA ENERGY LIMITED, BARIK BHAWAN, KOKATA-700072, FAX: 033-22360955
47. GENERAL MANAGER(O&M),BRBCL,NABINAGAR,BIHAR-824003,FAX-06332- 233026

CC:

Chief Engineer, OPM, CEA	Chief Engineer, NPC, CEA	ASSISTANT SECRETARY,ERPC
--------------------------	--------------------------	-----------------------------

ERPC:: Kolkata

पतों की सूची:

1. मुख्य अभियंता (ट्रांस., ओ एंड एम), बीएसपीटीसीएल, पटना, (फैक्स नं. 0612- 2504557/2504937)।
2. मुख्य अभियंता (सिस्टम ऑपरेशन), बीएसपीटीसीएल, पटना, (फैक्स नं. 0612- 2504557/2504937)।
3. मुख्य अभियंता, ट्रांसमिशन (ओ एंड एम), जेयूएसएनएल, रांची (फैक्स नं.-0651- 2490486/2490863)।
4. मुख्य अभियंता, टीवीएनएल, डोरंडा, रांची - 834102 (फैक्स नंबर 06544-225414)
5. मुख्य लोड डिस्पैचर, एसएलडीसी, ओपीटीसीएल, भुवनेश्वर (फैक्स नंबर 0674-2748509)
6. मुख्य महाप्रबंधक (ओ एंड एम), ओपीटीसीएल, भुवनेश्वर
7. एसआर. महाप्रबंधक (पीपी), ग्रिडको, जनपथ, भुवनेश्वर (0674-2547180)
8. निदेशक (संचालन), आईबी टीपीएस, एटी/पीओ बनहरपाली, झारसुगुड़ा, (फैक्स नंबर 06645-222225/222230)
9. महाप्रबंधक, टीटीपीएस, तालचेर, (फैक्स नंबर 06760-243212)
10. एसआर. महाप्रबंधक (विद्युत), ओएचपीसी लिमिटेड, भुवनेश्वर, (फैक्स नंबर 0674-2542102)
11. मुख्य अभियंता, सीएलडी, डब्ल्यूबीएसईटीसीएल, हावड़ा, (फैक्स नंबर 033-26886232)।
12. मुख्य अभियंता, केंद्रीय योजना विंग, डब्ल्यूबीएसईटीसीएल, साल्ट लेक (फैक्स नंबर: 033-23591955);
13. मुख्य अभियंता (पीटीआर), डब्ल्यूबीएसईडीसीएल, साल्ट लेक, कोलकाता (फैक्स:033-23345862)।
14. मुख्य महाप्रबंधक (ओएस), डब्ल्यूबीपीडीसीएल, कोलकाता-98 (फैक्स नंबर 033- 23393286/2335-0516)।
15. जीएम, कोलाघाट टीपीएस, डब्ल्यूबीपीडीसीएल, कोलाघाट (फैक्स नंबर 03228231280)
16. डीजीएम (ऑपरेशंस), डीपीएल, दुर्गापुर, (फैक्स नंबर 0343-2555052)
17. जीएम (एसवाईएस ऑपरेशन), सीईएससी, चौरंगी स्क्रायर, कोलकाता (फैक्स नंबर 033- 22253756/22129871)।
18. मुख्य अभियंता, एसएलडीसी, डीवीसी, हावड़ा (फैक्स नंबर 033-2688-5094)।
19. अपर मुख्य अभियंता, एसएलडीसी, विद्युत विभाग, शासन। सिक्किम, गंगटोक, (फैक्स नंबर 03592-228186/201148/202284)
20. कार्यकारी निदेशक, ईआरएलडीसी, पोसोको, कोलकाता, (फैक्स नंबर 033-2423-5809)
21. महाप्रबंधक, एफएसटीपीपी, एनटीपीसी, फरक्का, (फैक्स नंबर 03512- 224214/226085/226124)
22. महाप्रबंधक, खएसटीपीपी, एनटीपीसी, कहलगांव (फैक्स नंबर 06429-226082)
23. महाप्रबंधक, टीएसटीपीपी, एनटीपीसी, तालचेर, (फैक्स नंबर 06760-249053)
24. महाप्रबंधक (ओएस), पावरग्रिड, ईआर-II, कोलकाता (फैक्स नंबर: 033-23572827)
25. महाप्रबंधक, पावरग्रिड, ईआर-I, पटना, (फैक्स नं.0612-2531192)
26. महाप्रबंधक (ओ एंड एम), पावरग्रिड, ओडिशा प्रोजेक्ट्स, साहिद नगर, भुवनेश्वर - 751 007
27. प्रबंध निदेशक, डुक ग्रीन पावर कॉर्पोरेशन, पी.ओ. बॉक्स -1351, थिम्पस, भूटान - (फैक्स नंबर 00975-2336411)
28. प्रबंध निदेशक, भूटान पावर कॉर्पोरेशन, पी.ओ.
29. मुख्य अभियंता (ओ एंड एम), ताला एच.ई.प्रोजेक्ट, भूटान (फैक्स नंबर 009752/324803)
30. कार्यकारी निदेशक (ओ एंड एम), एनएचपीसी, फरीदाबाद (फैक्स नंबर:0129-2272413)
31. महाप्रबंधक, तीस्ता-वी पावर स्टेशन, एनएचपीसी, सिंगतम, पूर्वी सिक्किम (फैक्स 03592 - 247377)।
32. मुख्य अभियंता, रंगीत पावर स्टेशन, एनएचपीसी, पी.ओ. रंगीत नगर, दक्षिण सिक्किम (फैक्स नंबर 03595-259268)

33. वरिष्ठ उपाध्यक्ष, पीटीसी लिमिटेड, एनबीसीसी टावर्स, 15-भीकाजी काम प्लेस, नई दिल्ली-110066 (फैक्स नंबर 011-41659504)।
34. प्लांट हेड, आधुनिक पावर एवं नेचुरल रिसोर्सेज, झारखंड (फैक्स नं.: 0657-6628440)।
35. एजीएम (ऑपरेशंस), मैथन पावर लिमिटेड।
36. उपाध्यक्ष (विद्युत), वेदांता लिमिटेड, भुवनेश्वर- 751023 (फैक्स नंबर 0674-2302920)।
37. मुख्य विद्युत अभियंता, पूर्वी रेलवे, कोलकाता-700 001 (फैक्स नं.: 033-22300446)
38. मुख्य विद्युत अभियंता, दक्षिण पूर्व रेलवे, कोलकाता-43 (फैक्स: 033-24391566)।
39. उप निदेशक, पूर्वी आरपीएसओ, साल्ट लेक, कोलकाता- (फैक्स नं: 033- 23217075)
40. महाप्रबंधक (ओ एंड एम), एनएचपीसी लिमिटेड, फरीदाबाद, फैक्स: 0129-2272413
41. एसोसिएट वाइस प्रेसिडेंट, जीएमआर केईएल, भुवनेश्वर-751007। (फैक्स नंबर: 0674-2572794)
42. जीएम (एसओ एवं सीओएमएल), एनटीपीसी वीवीएनएल, नई दिल्ली-110033। फैक्स:011-24367021
43. श्री डी. पी. भागवा, मुख्य सलाहकार (ओ एंड एम), टेस्टा ऊर्जा लिमिटेड, नई दिल्ली-110 001 (फैक्स:011-46529744)।
44. श्री ब्रजेश कुमार पांडे, प्लांट हेड, जीतपीएल। (फैक्स:011-26139256-65)
45. निदेशक (एनपीसी), सीईए, एनआरपीसी बिल्डिंग, कटवारिया सराय, नई दिल्ली-110016
46. डीजीएम (ओएस), हल्दिया एनर्जी लिमिटेड, बारीक भवन, कोकाता-700072, फैक्स: 033-22360955
47. महाप्रबंधक (ओ एंड एम), बीआरबीसीएल, नबीनगर, बिहार-824003, फैक्स-06332- 233026

सीसी:

मुख्य अभियंता, ओपीएम, सीईए	मुख्य अभियंता, एनपीसी, सीईए	सहायक सचिव,ईआरपीसी
----------------------------	-----------------------------	--------------------

ईआरपीसी:: कोलकाता



MINUTES
OF
213TH OCC MEETING

Date: 14.03.2024

Eastern Regional Power Committee

14, Golf Club Road, Tollygunge

Kolkata: 700033

Contents

1. PART-A: CONFIRMATION OF MINUTES	3
1.1. Confirmation of Minutes of 212 th OCC Meeting held on 22 nd February 2024 virtually through Microsoft Teams online meeting platform	3
2. PART-B: ITEMS FOR DISCUSSION	3
2.1 Reliable Power supply of Kolkata and adjoining area: ERLDC	3
2.2 Update on installation of 7th (Interim) 500 MVA ICT at 400 kV Subhasgram (PG)-ERPC.....	5
2.3 Proposal for installation of SPS to prevent cascade tripping due to ICT tripping at Rajarhat(PG) :: West Bengal SLDC (WBSETCL)	7
2.4 Extension of timeline for completion of DTL from IBEUL to Sundergarh till September 2024 and Commissioning activities of Unit#2 using interim LILO arrangement -IBEUL	8
2.5 Submission of protection performance indices on monthly basis by users to RPC and RLDC for 220 kV and above lines- ERPC.	9
2.6 Long outage of 220 kV Talcher(NTPC)- Meeramundali-2 line: ERPC.....	10
2.7 Summer preparedness of ER: ERLDC.....	11
2.8 Information regarding updating of Restoration procedure of eastern region 2024: ERLDC	12
2.9 Planned Shutdown of NTPC Darlipalli Unit#1 in Apr 2024: NTPC Darlipalli	12
2.10 Difference in ABT(SCADA) vs SEM meters-NTPC Darlipalli.....	13
3. PART-C: ITEMS FOR UPDATE/FOLLOW-UP	14
3.1. ER Grid performance during February 2024.....	14
3.2. Ensuring the healthiness of ADMS	14
3.3. Commissioning status of ADMS.....	15
3.4. Primary frequency response of generating units in ER	15
4. PART-D: OPERATIONAL PLANNING	18
4.1. Anticipated power supply position during April-2024	18
4.2. Major Thermal Generating Units/Transmission Element outages/shutdown in ER Grid (as on 05-03-2024).....	18
4.3. Commissioning of new units and transmission elements in Eastern Grid in the month of February -2024.	23
4.4. UFR operation during the month of February 2024.	24

EASTERN REGIONAL POWER COMMITTEE

MINUTES OF 213TH OCC MEETING HELD ON 14.03.2024 (THURSDAY) AT 10:30 HRS

Member Secretary, ERPC chaired the 213th OCC meeting. On welcoming all the participants, he outlined the performance of ER grid during February 2024 and highlighted the following points:

- In February-2024, energy consumption of ER was **12901.67MU** which is **1.4 % more** than February-2023
- In February-2024, Peak demand met of ER was **23650 MW** which is **4.2% more** than February-2023..
- During February-2024, **74.1%** of time, the grid frequency was in IEGC Band (49.90Hz-50.05Hz).
- Thermal PLF of ER during February-2024 was **79 %**.
- Generating stations whose PLF was **more than 90%** during February-2024:

Utility	Generating station	PLF(%)
WBPDC	<i>Bandel TPS</i>	95
	<i>Bakreswar TPS</i>	95
	<i>Sagardighi TPS</i>	100
	<i>Santaldih TPS</i>	100
NTPC	<i>Darlipalli STPS</i>	98
	<i>Kahalgaon TPS</i>	91
	<i>Talcher stps</i>	92
DVC	<i>Bokaro TPS `A`</i>	94

- Transmission line (132 kV & above) commissioned during February- 2024:
NIL.
- **Coal stock position:**
 - There is marked improvement in coal stock position in Odisha(IB TPS),DVC and all NTPC generating units though it is still in sub-optimal range in some generating units especially Kolaghat TPS (WBPDC) having only 23% stock along with Barauni TPS (NTPC)(with 54% stock) and Mejia TPP(DVC)(with 48% stock) need to diligently focus on building their actual coal stock as per their normative requirement.
 - WBPDC was advised to stress on coal stock building through possibility of import for avoiding coal crisis during April(High demand period) and also to update status of imported coal(if any) till date.

- DVC was advised to sort out coal shortage issue at Mejia TPP thereby maintaining its coal stock at around 80% of normative requirement.
- Jharkhand (TVNL) was urged to enhance coal stock in face of impending demand surge and also advised to intimate constraints in building optimal coal stock despite having natural proximity to coalfields.
- Coal stock position (As on 11.03.2024) is detailed as follows:

SL.	Name of States/Power Stns.	% of Actual Stock vis-à-vis Normative Stock
1.	Jharkhand (TVNL)	51%
2.	Odisha/IBTPS	69%
3.	WBPDCCL	37 % (Min.Kolaghat TPS-23%, Max. Sagardighi TPS-40 %)
4.	DVC	77% (Min. Mejia TPP-48 %; Max Chandrapura TPS - 178%)
5.	NTPC	91% (Min Barauni TPS –54%; Max.North Karanpura STPS -174%)

- In this regard he highlighted that this is the most suited time to build coal stock keeping in view impending high demand season (Summer 2024) and thus advised all concerned GENCOs to strive for building optimum coal stock on sorting out various operational, commercial and administrative issues pertinent to coal linkage of respective generating units.
- He emphasized that all generating units having availed planned shutdown with consent of OCC in any of the previous OCC meetings must be restored to service as per the schedule approved by OCC forum.

ED, ERLDC at the outset emphasized following major issues of concern:

- Crunch high demand period in upcoming Summer amid general elections 2024.
- Earliest possible commissioning of North Karanpura TPP in this context, with its major share allocation to Jharkhand, was underlined.
- Challenges to be encountered in West Bengal network in upcoming Summer 2024 were underscored.

OPTCL representative affirmed of carrying out predictive maintenance of all transmission elements including protective relays and transformers ahead of upcoming high demand period.

JUSNL representative apprised of availing 170 MW share from North Karanpura generating station after COD while the power deficit shall be met from the market.

Bhutan representative apprised of the issue of erroneous as well as delayed receipt of operating codes for 400 KV lines from ERLDC as well as NLDC. ERLDC representative requested Bhutan to share exact dates and time of such occurrences for carrying out necessary remedial action.

GRIDCO representative highlighted:

Deficit of power owing to forced outage of Teesta-V(share of 114 MW) as well as less power availability from JITPL(60 MW instead of 120 MW)

West Bengal SLDC representative highlighted issues of power congestion in the 220 kV circuits from Malda(PG) through Gazole to Dalkhola(PG), driven by excessive load after meeting Gazole's requirements. The congestion was attributed to the near floating status of Kishanganj circuits and a consistent net export from Dalkhola towards Purnea, making this pattern problematic for system operations. To prevent cascade tripping, both the 220 kV Gazole-Dalkhola and the 132 kV Dalkhola-Gazole connections were disconnected, leading to a significant reduction in system redundancy, particularly concerning as summer approaches and in the light of upcoming elections. Despite requests to ERLDC for bus splitting at Dalkhola since January 10, 2024, no action has been taken, risking power outages in critical areas. The proposed solution includes bus splitting at Dalkhola or enhancing upstream contributions from Purnea/Kishanganj to relieve the Malda-Gazole segment. Failure to address the issue may necessitate the introduction of a Special Protection Scheme (SPS) in Gazole so that in case of flow beyond allowable limit in Malda(PG) - Gazole segment, to cut Gazole-Dalkhola D/C (if flow is from Gazole to Dalkhola that time) to reduce congestion in the remaining circuit of Malda(PG) to Gazole.

Bus splitting at 220 KV Dalkhola(PG) has already been carried out as confirmed by ERLDC representative.

1. PART-A: CONFIRMATION OF MINUTES

1.1. Confirmation of Minutes of 212th OCC Meeting held on 22nd February 2024 virtually through Microsoft Teams online meeting platform

The minutes of 212th Operation Coordination Sub-Committee meeting held on 22.02.2024 was circulated vide letter dated 04.03.2024.

Members may confirm the minutes of 212th OCC meeting.

Deliberation in meeting

Members confirmed the minutes of 212th OCC meeting.

2. PART-B: ITEMS FOR DISCUSSION

2.1 Reliable Power supply of Kolkata and adjoining area: ERLDC

Demand of Kolkata and the surrounding areas experiences a significant surge from mid-March onwards due to hot and humid weather conditions. Subhasgram S/S plays a crucial role in meeting a substantial portion of Kolkata and South 24 Parganas' power demand. Recent

developments indicate uncertainty regarding the commissioning of the 7th ICT of Subhasgram before the summer peak of 2024.

During Summer 2023, Subhasgram ICT loading exceeded 1800 MW, surpassing its rated capacity of 1760 MVA, resulting in a 200 MW load curtailment to adhere to the rated capacity. Anticipating load growth in West Bengal and potential delays in the installation of the 7th ICT, it is projected that Subhashgram ICTs may reach 2000 MW in Summer 2024, necessitating a load restriction of 400 MW during peak demand. Any tripping of ICTs could escalate the situation, leading to additional load shedding to prevent a blackout.

Rajarhat substation supports Kolkata's load, reaching 850 MW against its 1000 MW capacity in the previous summer. With closing of Bus Coupler at Newtown AAI, Rajarhat's ICTs loading is expected to approach its rated capacity.

To address these challenges, a necessary action plan must be formulated, including the implementation of a load trimming scheme, SPS, and other relevant measures to limit the loading of both substations.

ERLDC may update. Members may discuss.

Deliberation in meeting

*ERLDC representative delivered a comprehensive presentation (**Annexure B.2.1**) outlining the loading pattern along with capacity utilization in Subhasgram (PG) as well as Rajarhat(PG) S/S in last Summer(2023).*

Modalities of modified SPS at Subhasgram(PG) were highlighted as follows:

- In event of more than 110% loading at any ICT(1CT-3 or 4 or 5) along with tripping of any of the ICTs(315 MVA or 500 MVA), 200 MW load trimming shall take place at EMSS radial drawl point of CESC with preset time delay of 5 secs.*
- In case of more than 115% loading of any of the five ICTs at Subhasgram(PG), additional 150 MW load trimming at EMSS radial drawl point of CESC is proposed. This situation shall arise only in case of extreme exigency in event of SPS at Rajarhat(PG) fails to operate*

West Bengal SLDC representative submitted:

- Views of concerned beneficiaries must be considered before implementing additional SPS at Subhasgram(PG)*
- CESC load being connected in radial mode at Subhasgram(PG), there remains hardly any time to control power flow at this ISTS S/S manually by exchange of real time data with ERLDC.*
- Modalities of SPS as well as SOP on load trimming need to be finalized in a separate meeting.*
- Exploring the possibility of tap changing for optimal utilization of the 500 MVA ICT was proposed.*

Powergrid ER-II representative submitted:

- The 5th ICT(500 MVA) at Subhasgram(PG) had operated in Summer 2023 well beyond its safe temperature rise limits to cater to the unprecedented demand spike in Kolkata and*

adjoining areas. Continuous overloading with hardly any time for cooling had rendered this ICT highly vulnerable to failure during Summer of last year.

- The 5th ICT(500 MVA) at Subhasgram(PG) is already set at optimal tap position and Tap changing possibility cant be carried out as it would lead to circulating currents thereby adding to increased losses as well as further rise in WTI readings.

CESEC representative apprised:

- There was around 850 MW drawl by CESEC from Subhasgram(PG) during Summer 2023 with around 70-80 MW loadshedding in CESEC control area.
- More than 900 MW drawl is expected by CESEC at Subhasgram(PG) during Summer 2024.

OCC decision:

- OCC opined to convene a special meeting in physical mode on 20.03.2024 at Powergrid ER-II regional Headquarters comprising representatives from ERLDC,ERPC, Powergrid ER-II, West Bengal SLDC,WBSETCL,WBSEDCL and CESEC to further deliberate , reach at a consensus and finalize action plan on this critical issue.

2.2 Update on installation of 7th (Interim) 500 MVA ICT at 400 kV Subhasgram (PG)- ERPC.

As per the latest special weekly monitoring convened on 04.03.2024, Powergrid ER-II has confirmed the following:

Regarding Civil works:

- Powergrid ER-II representative delivered a brief presentation highlighting the latest status of activities duly substantiated with work site images.
- Foundation works for new 500 MVA ICT have been completed at Subhasgram (PG). Presently, only backfilling is under progress after completion of earthing works. RCC works of Sump pit wall in foundation area under progress that shall be completed in next 5-6 days.
- 220 kV CTA tower along with associated structures has been erected at Subhasgram (PG).

Regarding Electrical works:

- Necessary modification of 400 KV and 220 KV control and relay panel has been completed.
- RTCC panel along with bus reactor panel has been installed and powered up.
- All relevant tests on relays and associated panels have been carried out successfully.

ICT transportation Update:

- The 500 MVA ICT had reached Budge Budge on 16.02.2024 via waterways from Kolaghat.
- The ICT was rolled off on 20.02.2024.
- Upon mandatory checks of hydraulic system of the Trailer,the ICT was set for road transportation on 23.02.2024 near Achipur ghat, Budge Budge.

- Apprehension had been put forth by local municipality that newly constructed Charial bridge in the route of transportation may get damaged on failure to sustain the load of 500 MVA ICT.
- The bridge length being 30 meters and consignment spanning over 24 metres, entire load of the consignment shall be passed on the bridge at some instant while passing through.
- Local District Authorities and PWD (Govt of WB) thereby advised Powergrid to carry out some supportive strengthening activities at Charial Brige to withstand the load of 500 MVA ICT.
- Accordingly strengthening activities were commenced by Powergrid on 26.02.2024
- Subsequent to objection from local municipality authority, the strengthening activity was interrupted by local police on the night of 27.02.2024 and it is in halted state since then.
- The issue was thereafter communicated to Secretary(Power),Govt of WB. Subsequently Powergrid has been advised to carry out survey of alternate transportation route apart from Budge Budge with prudent consideration of all pros and cons and to submit a report on the same to Power dept, Govt of WB.
- Concern of keeping the ICT in dry air filled condition in unsupervised area for prolonged duration was also raised.
- Survey of Gardenreach jetty was carried out on 03.03.2024 and survey of the remaining options i.e Falta, Diamond Harbour and Khidirpore is to be completed by 05.03.2024. The survey is being carried out jointly by Powergrid and officials from Power dept, Govt. of WB
- Final communication encompassing alternate route survey details shall be made on behalf of Powergrid to Secretary(Power), Govt. of WB in a meeting tentatively scheduled on 06.03.2024.
- It was affirmed that notwithstanding the hindrance posed to ICT transportation, Subhasgram(PG) was made ready to commission the ICT by 20.03.2024.
- It was apprised that **availing alternate route of transportation** shall result in inevitable **time delay of minimum 3 to 4 months** as jetty construction needs to be done for roll-on and roll-off of the ICT.

Powergrid ER-II may update. Members may discuss.

Deliberation in meeting

Powergrid ER-II representative submitted:

- *All efforts have been put in by Powergrid by taking up strengthening works at Charial bridge on immediate basis as well as approaching respective State Departments (Govt. of West Bengal) to carry out the transportation of 500 MVA ICT to Subhasgram(PG) via the original route. However, pursuance with Govt of West Bengal yielding no positive outcome, the 500 MVA ICT transportation to Subhasgram(PG) S/S has got stalled in the midway.*
- *In continuation with deliberation at 5th special weekly monitoring on 04.03.2024, it was reiterated that availing alternate route of transportation shall result in inevitable time delay of minimum 3 to 4 months as jetty construction needs to be done for roll-on and roll-off of the ICT.*

- *Serious concern was raised on keeping the said 500 MVA ICT in dry condition for an indefinitely long duration that may adversely affect the healthiness of such a valuable asset, especially amid upcoming Norwester and monsoon season and unprotected Jetty Area at Budge Budge. So, if in any case no resolution fails to be undertaken, the said ICT may be required to be taken back to Regional Spare Pool.*

OCC decision:

- *OCC expressed grave concern on the present deadlock situation regarding 7th(interim) 500 MVA ICT transportation to Subhasgram(PG) and requested Powergrid ER-II to strive for resolving the intervening administrative issues hindering the ICT transportation via original selected route from Budge Budge.*

2.3 Proposal for installation of SPS to prevent cascade tripping due to ICT tripping at Rajarhat(PG) :: West Bengal SLDC (WBSETCL)

In view of the ensuing summer and Lok Sabha Election, all possible reviews of the system to find out the vulnerabilities and to safeguard the system against those efforts are ongoing. As a result of it, the apprehended peak loading of Rajarhat(PG) sub-station is found to be around 826 MW (April'24). With a peak load around 800 MW, if one ICT at Rajarhat(PG) sub-stn trips, a huge part of North 24 Parganas, Kolkata and South 24 parganas power supply may be interrupted suddenly for cascade tripping. To avoid this type of grid hazard with dark out for a massive area with conductor snapping even is extremely undesirable.

Hence, it is proposed to introduce a **SPS (special protection scheme) which will make instant tripping of 220 KV Rajarhat-New Town AA3 D/C from Rajarhat(PG) end along with tripping of Subhasgram(PG) -New Town AA3 circuit and Subhasgram(PG) - KLC circuit at Subhasgram(PG) end using OPGW connectivity between Rajarhat and Subhasgram(PG) Substations.** This will ensure non propagating the impact of one transformer tripping at Rajarhat(PG) substation up to Subhasgram(PG) end and will save the system of Subhasgram (PG), Kasba (WBSETCL) point (synchronising point for CESC) etc. Also the 220 kV Subhasgram(PG) -New Town AA3 circuit and Subhasgram(PG) - KLC circuit will be saved from any possibility of conductor snapping. Power Grid may please be requested to explore the implementation of the said SPS before forthcoming election.

In 212th OCC meeting,

- OCC advised ERLDC to carry out further comprehensive study assessing requirement of proposed SPS by WB SLDC taking other essential eastern regional grid parameters into consideration.
- OCC advised West Bengal SLDC to share the necessary details to carry out the study of SPS with ERLDC at earliest.

West Bengal SLDC (WBSETCL) and ERLDC may update. Members may discuss.

Deliberation in meeting

Modalities of SPS at Rajarhat(PG) were highlighted by ERLDC as follows:

- In event of more than 110% loading at any ICT along with tripping of any of the ICTs, tripping 220 KV Rajarhat-AAII, 220 KV Rajarhat-AAII will take place & SPS signal will be sent to Subhasgram(PG) where Tripping of 220 KV Subhasgram-KLC- AAIII line & 220 KV Subhasgram-AAIII line shall be carried out.
- This SPS scheme is proposed to ensure confinement of such grid disturbance locally to North 24 parganas rather than propagating to south 24 parganas and most importantly to Kasba(WB) S/S, the synchronizing point of CESC.

Powergrid ER-II representative submitted that healthiness of communication equipment (DTPC) should be ensured for proper functioning of the proposed SPS as considerable delay in signal propagation from Rajarhat(PG) to Subhasgram(PG) may defeat the purpose of SPS .

It was also emphasized that there is practically no room available in 500 MVA ICTs at Rajarhat(PG) for additional load sharing. The apparent scope available for load sharing in 500 MVA ICTs at Rajarhat(PG) is actually the difference WTI and OTI readings, thus additional load sharing may violate safety guidelines for operating an ICT at 400 kV.

West Bengal SLDC representative submitted:

For coordinated operation of SPS at both Rajarhat(PG) and Subhasgram(PG) shall require rearrangement of outgoing feeders with proper decoupling of load using bus coupler so that load doesn't get transferred suddenly to Subhasgram(PG) immediately after SPS operation at Rajarhat(PG).

OCC decision:

- OCC opined to convene a special meeting in physical mode on 20.03.2024 at Powergrid ER-II regional Headquarters comprising representatives from ERLDC,ERPC, Powergrid ER-II, West Bengal SLDC,WBSETCL,WBSEDCL and CESC to further deliberate and forge a consensus on the modalities of SPS installation at 400 kV Subhasgram(PG) and 400 kV Rajarhat(PG) S/S.

2.4 Extension of timeline for completion of DTL from IBEUL to Sundergarh till September 2024 and Commissioning activities of Unit#2 using interim LILO arrangement -IBEUL

IBEUL has synchronized U#1 of 350 MW on 13th January 2024, on interim arrangement that is LILO on IB Thermal Power station to PGCIL Sundergarh 400 kV line.

LILO was charged on 8th January 2024 and completion of direct transmission line expected by April-2024, progress report of the same is being communicated to all concerned.

The Director, Power System Project Monitoring Division(CEA) opined that due to increasing demand & capacity addition targets proposed by MOP, commissioning of Unit#2 at the earliest is crucial and thus after due examination by CTUIL, the commissioning activities of Unit#2 may be considered using LILO arrangement. Further, the timeline of completion of DTL may be extended till September 2024.

Relevant communication from CEA attached at **Annexure B.2.4.**

IBEUL may update. Members may discuss.

Deliberation in meeting

IBEUL representative informed:

- DTL from IBEUL to Sundergarh shall be ready for commissioning by middle of April 2024.
- In respect of Unit#02(350 MW), Hydro test of the boiler is completed, all associated Boiler Feed pumps ready to be put in service and boiler lightup to be done by 28.03.2024. Barring gear of turbine to be tested and installed by 24.03.2024.
- The Unit#02 to be synchronized by end of April 2024.
- Connectivity of the to be commissioned Unit#02(350 MW) has already been applied to CTU.

OCC decision:

OCC advised IBEUL to strictly adhere to the submitted timelines and also regularly update the status of both DTL and Unit#02(350 MW) commissioning for monitoring on continuous basis.

2.5 Submission of protection performance indices on monthly basis by users to RPC and RLDC for 220 kV and above lines- ERPC.

Users shall submit the following protection performance indices of previous month to their respective RPC and RLDC on monthly basis for 220 kV and above (132 kV and above in NER) system, which shall be reviewed by the RPC:

(a) The Dependability Index defined as $D = \frac{N_c}{N_c + N_f}$

where,

N_c is the number of correct operations at internal power system faults and

N_f is the number of failures to operate at internal power system faults.

(b) The Security Index defined as $S = \frac{N_c}{N_c + N_u}$

Where,

N_c is the number of correct operations at internal power system faults

N_u is the number of unwanted operations.

(c) The Reliability Index defined as $R = \frac{N_c}{N_c + N_i}$

Where,

N_c is the number of correct operations at internal power system faults

N_i is the number of incorrect operations and is the sum of N_f and N_u

Further, as per IEGC 2023 Clause 15 (7),

"Each user shall also submit the reasons for performance indices less than unity of individual element wise protection system to the respective RPC and action plan for corrective measures. The action plan will be followed up regularly in the respective RPC."

In 131st PCC meeting, all utilities were advised to submit mentioned protection performance indices of 220 kV and above system (132 kV and above for Sikkim) to ERPC/ERLDC every month in compliance to the Grid Code.

In 133rd PCC meeting, representatives of state utilities opined that respective SLDCs may be approached for the submission of above data. After deliberation, PCC observed that SLDCs being the nodal agency in intra-state system, the responsibility for submission of the protection performance indices shall be with them. PCC advised all concerned SLDCs to ensure timely submission of the above information to ERLDC/ERPC.

As representatives of SLDCs (except SLDC Odisha) are not present in the meeting, it was decided to refer the issue to OCC meeting for discussion and compliance.

SLDCs may update. Members may discuss.

Deliberation in meeting

West Bengal SLDC representative submitted:

- *Supervising protection of intra-state system is done by STU, so it will be convenient to collect the relevant data from STU rather than SLDC.*
- *The required details of protection performance need to be substantiated with relevant fault analysis which is carried out by testing department of WBSETCL and not at WB SLDC.*

SLDC Odisha representative asserted that all protection related studies and planning carried out by STU while SLDC is concerned only with the operational aspect. Similar views expressed by Jharkhand SLDC.

SLDC Bihar representative was not present in the meeting.

DVC SLDC acknowledged the role of SLDC as nodal agency in intra-state system and affirmed to submit the protection performance indices to ERPC and ERLDC.

OCC decision:

- *OCC suggested all SLDCs to leverage State level Coordination forums for gathering relevant data of protection performance from the respective constituents in convenient way.*
- *OCC also opined that as per IEGC 2023 the details of protection performance to be submitted by all users including all generating utilities including CPPs, all transmission licensees and DISCOMs, hence SLDCs as nodal agency in intra-state transmission system shall be entrusted with the responsibility of submitting necessary protection performance details of all constituents of their respective states to ERPC and ERLDC.*

2.6 Long outage of 220 kV Talcher(NTPC)- Meeramundali-2 line: ERPC

220kV Talcher (NTPC) -Meeramundali -D/C tripped on 10.06.2023 due to Tower collapse at loc no 41, 42 (from Meeramundali end). 220kV Talcher (NTPC) -Meeramundali -1 was revived on 29.06.2023 with ERS Tower and 220kV Talcher (NTPC) -Meeramundali -2 is still under breakdown condition.

220kV Talcher (NTPC) -Meeramundali -D/C is an important Intra-State line between Talcher (NTPC) and Meeramundali. Long outage of this lines impacts the system reliability.

Considering the above and keeping in view ensuing summer, cooperation is solicited to expedite the revival of 220kV Talcher (NTPC) -Meeramundali -2 at the earliest.

GRIDCO may Update the status and share the charging plan of ckt-2.

Members may discuss.

Deliberation in meeting

OPTCL representative submitted:

- *Encountering severe ROW issues has led to such inordinate delay in restoration of 220 kV Talcher(NTPC)- Meeramundali-ckt-2 which is under long outage since 10.06.2023.*
- *220 kV Talcher(NTPC)- Meeramundali-ckt-2 shall be restored to service by March'2024.*

OCC decision:

OCC advised OPTCL to restore 220 kV Talcher(NTPC)- Meeramundali-ckt-2 as per the submitted timeline.

2.7 Summer preparedness of ER: ERLDC

As the summer season approaches, the country is preparing for a surge in demand due to the increased of space cooling loads. Eastern Region characterized by hot and humid climates, experiences peak demand during the summer. To ensure adequate power generation and transmission/distribution infrastructure is in place to meet the expected demand and minimize the risk of service disruptions detailed threadbare discussion among all the beneficiaries is required.

In view of this, ERLDC will conduct a meeting on Summer Preparedness through video conference mode on 21st March'2024. All the SLDCs, regional generating companies, State embedded generating companies, Power management or procurement cell of Discoms & ALDCs are requested to join the meeting. Agenda points is appended in the **Annexure B.2.6**.

Members may discuss.

Deliberation in meeting

ERLDC representative apprised of organizing Summer Preparedness meeting on 28.03.2024 via virtual mode in a bid to address the pressing challenges amid predicted demand escalation and to carry out proper planning for ensuring reliable grid operation.

OCC decision:

OCC, on commending such initiative from ERLDC, advised all concerned SLDCs, eastern regional generating companies (Central sector and IPPs), State embedded generating companies, Power management or procurement cell of DISCOMS & ALDCs to actively participate in the Summer Preparedness meeting on 28.03.2024 via virtual mode and to share their comprehensive plan in regard to efficient demand management in ER grid in upcoming Summer 2024.

2.8 Information regarding updating of Restoration procedure of eastern region 2024: ERLDC

Black Start and Restoration Procedure of Eastern Region for 2024 was circulated via email on 28th December 2023, to all constituents, ISGS, IPP, ISTS Licensees, and all divisions of Railway operating in the Eastern region, where they were requested to kindly review the same and give their valuable feedback so that the document may be finalized by the 13th of January 2024.

Based on further inputs received from the utilities/stakeholders of the Eastern region up to 27th Jan 2024, the Black Start and Restoration procedure of the Eastern Region has been finalized and considered to be approved, pursuant to regulation No. 34 (1), 34(2) of the IEGC. The final version of the document was also shared with all constituents/ISGS/IPP/ISTS Licensees on 07th February 2024.

ERLDC may update. Members may discuss.

Deliberation in meeting

ERLDC representative informed of finalization of black start and Restoration procedure of Eastern Region in line with IEGC 2023.

OCC acknowledged the same.

ADDITIONAL AGENDA

2.9 Planned Shutdown of NTPC Darlipalli Unit#1 in Apr 2024: NTPC Darlipalli

- Unit #1 is running with very low LP differential expansion. LP diff value in DCS is not reflecting actual running clearance between rotor and station diaphragm.
- Already two LP turbines got damaged due to low LP differential expansion. Spare LP turbine is not available with NTPC.
- In view of above, to take preventive action and to avoid any damages to LPT rotor, Unit#1 must be taken early shutdown to check actual clearances and remedial measures during overhauling.
- It is requested to approve the Unit # 1 planned shutdown in the month of April 2024. Final date shall be confirmed after mobilization of man and material.

NTPC Darlipalli may update. Members may discuss.

Deliberation in meeting

NTPC Darlipalli representative submitted :

- *Damage caused to LP turbines due to differential expansion and non-availability of spare LP turbine at NTPC Darlipalli.*
- *Exact date of availing shutdown in April 2024 is not yet finalized. It may be availed in April 2024 on recommendation of concerned OEM of turbine.*

OCC decision:

OCC opined that any proposal of planned shutdown in April 2024 has to be submitted to CEA from NTPC Corporate Office as no planned shutdown of generating units in April 2024 shall be granted consent from OCC forum in line with MOP guidelines.

2.10 Difference in ABT(SCADA) vs SEM meters-NTPC Darlipalli

- In 212th OCC meeting, "OCC suggested NTPC Darlipalli to recheck accuracy of associated CT/CVT as well as other electrical parameters of the main and check meters(Genus make) to figure out the root cause for mismatch in reading. Highlighting the issue to concerned OEM of the meters"
- CT/CVT is common for Main, Check and SCADA meters
- As discussed in last OCC, error between Main and Check meter to be taken up with meter OEM by PGCIL

NTPC Darlipalli may update. Members may discuss.

Deliberation in meeting**OCC decision:**

- *OCC suggested NTPC Darlipalli to replace the existing meters (SEM and SCADA meters) with different serial numbers of the same OEM in coordination with the concerned OEM and Powergrid.*
- *OCC also advised NTPC Darlipalli to carry out further root cause analysis of mismatch in SEM vs SCADA meter data with aid of relevant data shared by ERLDC.*

3. PART-C: ITEMS FOR UPDATE/FOLLOW-UP

3.1. ER Grid performance during February 2024.

The average and maximum consumption of Eastern Region and Max/Min Demand (MW), Energy Export for the month February -2024 were as follows:

AVERAGE CONSUMPTION (MU)	MAXIMUM CONSUMPTION(MU)/ DATE	MAXIMUM DEMAND (MW)	MINIMUM DEMAND (MW)	SCHEDULE EXPORT	ACTUAL EXPORT
		DATE/TIME	DATE/TIME	(MU)	(MU)
447.2 MU	468.4 MU 29.02.2024	23650 MW, 24.02.2024 at 18:53 Hrs.	14558 MW, 05.02.2024 at 04:09 Hrs.	4965	5424

ERLDC/ERPC may highlight the performance of the ER grid.

Deliberation in meeting

The grid performance of ER for the month of February -2024 was highlighted.

3.2. Ensuring the healthiness of ADMS

State	Criteria for ADMS operation	Number of instances for which ADMS criteria satisfied	Number of instances for which detail received	Discussion regarding previous month performance	Update in 212 th OCC meeting
West Bengal	1. System Frequency < 49.7 Hz 2. WB over-drawl > 150 MW 3. Delay = 4 min	0	-	-	-
Jharkhand	1. System Frequency < 49.9 Hz 2. Jharkahnd over-drawl > 25 MW 3. Delay = 3 min	149	Nil	-	-

DVC	1. System Frequency < 49.9 Hz 2. DVC over-drawl > 150 MW 3. Delay = 3 min	9	Nil	-	-
Odisha	1. System Frequency < 49.9 Hz 2. Odisha over-drawl > 150 MW 3. Delay = 3 min	3	Nil	-	-

Members may note.

Deliberation in meeting

Members noted.

3.3. Commissioning status of ADMS

Automatic demand management scheme (ADMS) has been already commissioned in West Bengal, DVC, Odisha and Jharkhand. However, for Bihar it is yet to be implemented, the last status as confirmed in the earlier meeting is as follows.

SI No	State/Utility	Logic for ADMS operation	Target Date
1	Bihar	F < 49.7 AND deviation > 12 % or 150 MW	

Bihar may update the status of the implementation of ADMS scheme.

Members may note.

Deliberation in meeting

Members noted.

3.4. Primary frequency response of generating units in ER

Adhering to IEGC clauses 30.8 and 30.10.(a) to 30.10.(q), generating stations within the eastern region must furnish essential data to ERLDC within two days of receiving a notification regarding a reportable frequency event. Furthermore, according to clause 30.10.(n), all control areas within

the eastern region are obligated to evaluate their frequency response characteristics and share the assessment, along with high-resolution data, with ERLDC. Following each reportable frequency event, ERLDC disseminates a format to entities, facilitating the collection of necessary data for their performance assessment. Improper/non-receipt of any data compels ERLDC to utilize Scada data for calculating the performance of the respective control area. Further based on this Scada data decision for retesting of primary frequency response of concerned generators will be done. Therefore, timely submission of primary response data is crucial to comply with IEGC.

The plant wise data submission statistics for frequency event flagged by ERLDC up to 01.03.2024 is given below:

STATIONS		17.12.2023	05.01.2024	15.01.2024	15.01.2024
		13:01	05:16	13:59	14:06
ADHUNIK		Received	Received	Received	Received
BARH	ISGS	Pending	Received	Received	Received
BRBCL	ISGS	Pending	Pending	Pending	Pending
DARLIPALLI	ISGS	Received	Received	Received	Received
DIKCHU		PLANT OUT	PLANT OUT	PLANT OUT	PLANT OUT
TASHIDING		Pending	Pending	Pending	Pending
FARAKKA	ISGS	Pending	Pending	Pending	Pending
GMR		Received	Received	Received	Received
JITPL		Received	Received	Received	Received
KAHALGAON	ISGS	Received	Received	Received	Received
MPL		Received	Received	Received	Received
NPGC	ISGS	Received	Received	Received	Received
TALCHER	STG-1&2 ISGS	Received	Received	Received	Received
TEESTA III	ISGS	PLANT OUT	PLANT OUT	PLANT OUT	PLANT OUT
TEESTA V	ISGS	PLANT OUT	PLANT OUT	PLANT OUT	PLANT OUT
North Karanpura	ISGS	Pending	Pending	Pending	Pending

In view of the same all utilities are once again requested to kindly look into the matter and take necessary action to ensure consistent.



= Data received

Data submission for every frequency event flagged by ERLDC.



= Data not received

Members may note.

Deliberation in meeting

Members noted.

4. PART-D: OPERATIONAL PLANNING

4.1. Anticipated power supply position during April-2024

The abstract of peak demand (MW) vis-à-vis availability and energy requirement vis-à-vis availability (MU) for the month of April 2024 were prepared by ERPC Secretariat (**Annexure D.1**) on the basis of LGBR for 2023-24 and feedback of constituents, keeping in view that the units are available for generation and expected load growth etc.

Members may update.

Deliberation in meeting

*The updated anticipated power supply position for April 2024 is provided at **Annexure D.1***

4.2. Major Thermal Generating Units/Transmission Element outages/shutdown in ER Grid (as on 05-03-2024)

a) **Thermal Generating Stations outage report:**

SL No	STATION	STATE	AGENCY	UNIT NO	CAPACITY (MW)	REASON(S)	OUTAGE DATE
1	BARAUNI TPS	BIHAR	NTPC	7	110	Poor condenser vacuum	19-Jul-2023
2	BARAUNI TPS	BIHAR	NTPC	6	110	Low vacuum	22-Jul-2023
3	JITPL	ODISHA	JITPL	1	600	IP control valve close, turbine blade got cracked	08-Feb-2024
4	Sterlite	ODISHA	SEL	3	600	Annual Overhauling and R&M activity	13-Feb-2024
5	RTPS	DVC	DVC	2	600	DVC informed due to very low lube oil pressure, the unit got forced tripped on 26.02.2024 and since the unit was approved planned S/d for 28 days w.ef 28.02.2024 for FGD,De-NOx and pollution equipment upgradation works , the planned S/D of the	26-Feb-2024

						unit was availed for 30 days w.e.f 00.00 hrs of 27.02.2024	
6	OPGC	ODISHA	OPGC	4	660	Problem in Generator AVR	01-Mar-2024
7	MEJIA TPS	DVC	DVC	1	210	Due to suspected BTL in water wall zone	04-Mar-2024
8	FSTPP	WEST BENGAL	NTPC	5	500	Drum level high	05-Mar-2024

All Generating stations are requested to update expected restoration time and reason outage to ERLDC/ERPC on weekly basis in case of any change at their end.

b) Major Generating stations Out on Reserve Shutdown due to low system demand:

SL No	STATION	STATE	AGENCY	UNIT NO	CAPACITY (MW)	REASON(S)	OUTAGE DATE
1	SOUTHERN	WEST BENGAL	CESC	1	67.5	Low System Demand	21-Feb-2024
2	SOUTHERN	WEST BENGAL	CESC	2	67.5	Low System Demand	09-Feb-2024
3	NABINAGAR(NPGC)	BIHAR	NTPC	2	660	Low system demand.	02-Mar-2024

c) Hydro Unit Outage Report:

S. NO	STATION	STATE	AGENCY	UNIT NO	CAPACITY (MW)	REASON(S)	OUTAGE DATE
1	TEESTA STG III Hep	SIKKIM	TUL	1	200	Sudden cloudburst at glacier fed LOHNAK Lake followed by huge inrush of water in Teesta River and damage of Teesta III Dam & downstream Powerhouses	04-Oct-2023
2	TEESTA STG III Hep	SIKKIM	TUL	2	200		
3	TEESTA STG III Hep	SIKKIM	TUL	3	200		
4	TEESTA STG III Hep	SIKKIM	TUL	4	200		

5	TEESTA STG III Hep	SIKKIM	TUL	5	200		
6	TEESTA STG III Hep	SIKKIM	TUL	6	200		
7	DIKCHU Hep	SIKKIM	SKPPL	1	48	Sudden cloudburst at glacier fed LOHNAK Lake followed by huge inrush of water in Teesta River and damage of Teesta III Dam & downstream Powerhouses	04-Oct-2023
8	DIKCHU Hep	SIKKIM	SKPPL	2	48		
9	TEESTA HPS	SIKKIM	NHPC	1	170	Sudden cloudburst at glacier fed LOHNAK Lake followed by huge inrush of water in Teesta River and damage of Teesta III Dam & downstream Powerhouses	04-Oct-2023
10	TEESTA HPS	SIKKIM	NHPC	2	170		
11	TEESTA HPS	SIKKIM	NHPC	3	170		
12	BALIMELA HPS	ODISHA	OHPC	3	60	The unit taken out under R&M since 08/07/2022 for 18 months.	08-Jul-2022
13	INDRAVATI	ODISHA	OHPC	2	150	Capital Maintenance	23-Nov-2023
14	CHIPLIMA HPS / HIRAKUD II	ODISHA	OHPC	1	24	Capital Overhauling	15-Dec-2023
15	RANGIT HPS	SIKKIM	NHPC	1	20	Capital maintenance	02-Jan-2024
16	RANGIT HPS	SIKKIM	NHPC	3	20	Under Water parts Checking	27-Feb-2024
17	JORETHANG	SIKKIM	DANS	1	48	Barrage Civil Work	08-Jan-2024
18	JORETHANG	SIKKIM	DANS	2	48	Barrage Civil Work	16-Jan-2024
19	CHUZACHEN	SIKKIM	GATI	1	55	Annual Overhauling	31-Jan-2024

20	BURLA HPS/HIRAKUD I	ODISHA	OHPC	7	37.5	Annual maintenance	13-Feb-2024
21	RENGALI HPS	ODISHA	OHPC	3	50	Annual maintenance	07-Feb-2024

d) Long outage report of transmission lines (As on 10.03.2024):

Transmission Element / ICT	Outage From	Reasons for Outage
220/132KV 100 MVA ICT II AT LALMATIA	22.01.2019	Commissioning work of 220/132KV, 100MVA Transformer and its associated control Panel under progress.
220 KV PANDIABILI - SAMANGARA D/C	03.05.2019	Tower Collapsed during Cyclone FANI (Restoration project is entrusted upon PGCIL & 220kV Samangara-Pandiabili ckt-I&II are anti-theft charged from Pandiabili end from loc no.01 to loc no.74)
220/132KV 100 MVA ICT 3 AT CHANDIL	30.04.2020	Due to Fire hazard ICT damaged and burnt.
400/220KV 315 MVA ICT 4 AT JEERAT	09.04.2021	Due to Fire hazard ICT damaged and burnt. It was decided that 315MVA, 400/220KV spare ICT from Regional Pool at Malda to be replaced in place of defective 315 MVA ICT (4th) at Jeerat 400 KV S/S(WB). Work under progress.
220KV-FSTPP-LALMATIA-I	21.04.2021	Conductor stringing 12.965 km has been completed and Stringing between Tower Loc. no. 152 to 159 is under progress. Transmission line is idle charged between Lalmatia GSS end to Tower Loc.no.169
220KV-MUZAFFARPUR(PG)-GORAUL(BH)-1	11.06.2022	Main Bay is under breakdown due to flashing in GIS module
220KV-WARIA-BIDHANNAGAR-1 & 2	08.06.2022	To control overloading of 220 kV Waria-DSTPS (Andal) D/C line
400/220KV 315 MVA ICT 2 AT PATRATU	27.09.2022	ICT tripped on few occasions due to Buchholz later DGA violation found, internal fault in transformer to be rectified. (DGA violation)
132KV-BARHI-RAJGIR-1	25.03.2023	Dismantling of tower no. 227, 228, and 229 crossing the premises of

132KV-NALANDA-BARHI(DVC)-1	25.03.2023	Mahabodhi Cultural centre along with Destraining of conductor of both circuits and Earth wire between tension tower no. 218-237 in same line.
220KV-TSTPP-MEERAMUNDALI-2	10.06.2023	Tower collapse at loc no 41, 42 (from Meramundali end). Ckt1 charged through ERS.
400KV-RANGPO-TEESTA-V-1 & 2	04.10.2023	Tower near gantry of Teesta V powerhouse collapsed due to sudden cloudburst at glacier fed LOHNAK Lake followed by huge inrush of water in TEESTA river and damage of Teesta III Dam & downstream Powerhouses
400KV-TEESTA-III-RANGPO-1	04.10.2023	Hand tripped from Teesta-III end due to sudden cloudburst at glacier fed LOHNAK Lake followed by huge inrush of water in TEESTA river and damage of Teesta III Dam & downstream Powerhouses
400KV-TEESTA-III-DIKCHU-1	04.10.2023	
400KV-RANGPO-DIKCHU-1	04.10.2023	Hand tripped from Rangpo end due to sudden cloudburst at glacier fed LOHNAK Lake followed by huge inrush of water in TEESTA river and damage of Teesta III Dam & downstream Powerhouses
400KV JHARSUGUDA-ROURKELA-1 &3	26.10.2023	Reconductoring work
220KV/132KV 160 MVA ICT 2 AT MALDA	30.11.2023	For shifting of the ICT- 2 from 132KV AIS to GIS
400KV-BINAGURI-BONGAIGAON-1	29.01.2024	Reconductoring work of 400 KV Bongaigaon Binaguri-1
400KV-BINAGURI-TALA-2	31.01.2024	Annual maintenance at Tala end
400KV/220KV 315 MVA ICT 1 AT FARAKKA	21.02.2024	Replacement of 220kV circuit breaker of 315 MVA ICT-1 bay under ADDCAP.
400KV/220KV 315 MVA ICT 2 AT TSL KALINGANAGAR	22.02.2024	RTV Coating, GIS Flange greasing of ICTs
400KV-KHSTPP-BANKA (PG)-1	24.02.2024	Switchyard bay updation work

Transmission licensees/ Utilities are requested to update expected restoration date & work progress regarding restoration regularly to ERLDC/ERPC on monthly basis by 5th of each month so that status of restoration can be reviewed in OCC. Utilities are also requested to update outage of any elements within their substation premises like isolator/breaker to ERLDC/ERPC regularly. (Reported as per Clause 5.2(e) of IEGC)

Deliberation in meeting

Members noted.

4.3. Commissioning of new units and transmission elements in Eastern Grid in the month of February -2024.

The details of new units/transmission elements commissioned in the month of February-2024 based on the inputs received from beneficiaries:

NEW ELEMENTS COMMISSIONED DURING FEBRUARY, 2024							
GENERATING UNITS							
SL. NO.	Location	Owner/ Unit name	Unit No / Source	Capacity added (MW)	Total/Installed Capacity (MW)	DATE	Remarks
NIL							
ICTs/ GTs / STs							
SL. NO.	Agency/ Owner	SUB-STATION	ICT NO	Voltage Level (kV)	CAPACITY (MVA)	DATE	Remarks
NIL							
TRANSMISSION LINES							
SL. NO.	Agency/ Owner	Line Name		Length (KM)	Conductor Type	DATE	Remarks
NIL							
LILO/RE-ARRANGEMENT OF TRANSMISSION LINES							
SL. NO.	Agency/ Owner	Line Name/LILO at		Length (KM)	Conductor Type	DATE	Remarks
1	PGCIL	400KV-PATNA-BALIA-3		180	Quad Moose	06-02-2024	400kV Balia(PG) - Patna(PG) Ckt 3 (After bypassing 400kV Balia(PG) - Naubatpur(BS) Ckt 1 & Patna(PG) - Naubatpur(BS) Ckt 1 outside Naubatpur(BS))
BUS/LINE REACTORS							
SL. NO.	Agency/	Element Name		SUB-STATION	Voltage	DATE	Remarks

	Owner			Level (kV)		
NIL						
BUS						
SL. NO.	Agency/ Owner	Element Name	SUB-STATION	Voltage Level (kV)	DATE	Remarks
NIL						
BAYS						
SL. NO.	Agency/ Owner	Element Name	SUB-STATION	Voltage Level (kV)	DATE	Remarks
1	NTPC NKSTPP	400KV MAIN BAY OF ST-2 AT 400KV NORTH KARANPURA SS	NORTH KARANPURA SS	400	09-02-2024	NTPC NKSTPP ST#2 was through the Tie Bay in Dec 2023.

Members may note.

Deliberation in meeting

Members noted.

4.4. UFR operation during the month of February 2024.

Frequency profile for the month as follows:

MONTH	MAX	MIN	% LESS IEGC BAND	% WITHIN IEGC BAND	% MORE IEGC BAND
	(DATE/TIME)	(DATE/TIME)			
Feb, 2024	50.35 Hz on 01-02-2024 at 06:03 hrs	49.57 Hz on 18-02-2024 at 09:18 hrs	6.24	74.07	19.69

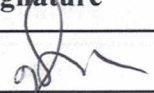
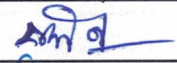
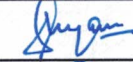
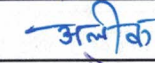
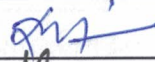
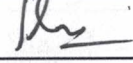
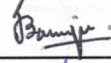
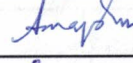
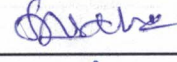
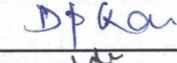
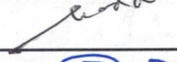
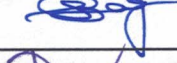
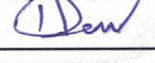
Hence, no report of operation of UFR has been received from any of the constituents.

Members may note.

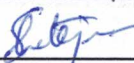
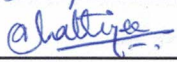
Deliberation in meeting

Members noted.

Annexure-A: List of Participants in 213th OCC meeting

Sl. No.	Name	Designation	Organisation	Contact No.	E-mail Id	Signature
1	N.S.Mondal	Member Secretary	ERPC	9958389967	mserpc-power@gov.in	
2	R. Sutradhar	Executive Director	ERLDC	9436302714	rajibsutradhar@posoco.in	
3	S. KESRIWAL	SE	ERPC	9831919509	shyam.kesriwal@gov.in	
4	A. De	DD	ERPC	9681932906	alix.erpc@gov.in	
5	P.K.DE	SE	ERPC	9433125844	prasadde.cea@gov.in	
6	Raju Kachhap	Sr. Manager	SLDC JUSNL	7783087568	rajmaifme82@gmail.com	
7	Preetam Banerjee	Addl. C.E	WBSEDC	7003871189	preetan72@gmail.com	
8	Rita Chakrabarty	CE	SLDCWB	9434910030	ce.wbstdc@gmail.com	
9	Shouvik Banerjee	ACE, SLDC	WBSETCL	9434910379	svkbanerjee@yahoo.com	
10	Apn Mazumder	GM(OS)	WBPDCL	8336903770	amazumder@wbpdcl.co.in	
11	S.M.S. Sahoo	DGM (EI.)	OPTCL	9438908353	ele.smsahoo@optcl.co.in	
12	Sanjaya Kumar Das	SrGM (PP)	SLDC ODISHA	9438907414	sanjayasldc@gmail.com	
13	Bijay Kumar Das	SrGM (PP)	GRIDCO ODISHA	9438907699	Elc.bkdas@gridco.co.in	
14	A. K. Mohanty	SrGM, OHPC	OHPC	7326840019	akm_678@yahoo.co.in	
15	Debarshi De	SM, (500)	CEEC	9230521123	debarshi.de@npsc.in	
16	S. K. Maharana	DGM (Elec.)	GRIDCO	9438591100	ele.skmaharana@gridco.co.in	
17	Debi Prasad Kar	DGM (Elect.)	SLDC, ODISHA	9438907416	ele.dpkar@sldc odisha.org.in	
18	Mamuj Podder	AGM (ELECT)	WBPDCL	8336904077	mpodder@wbpdcl.co.in	
19	S. K. BAG.	Addl. C.E.	SLDC, WB	7980098826	sayalkbag74@gmail.com	
20	P. Sen	Manager (OPN)	DPL	8013843947	palash_239@rediffmail.com	

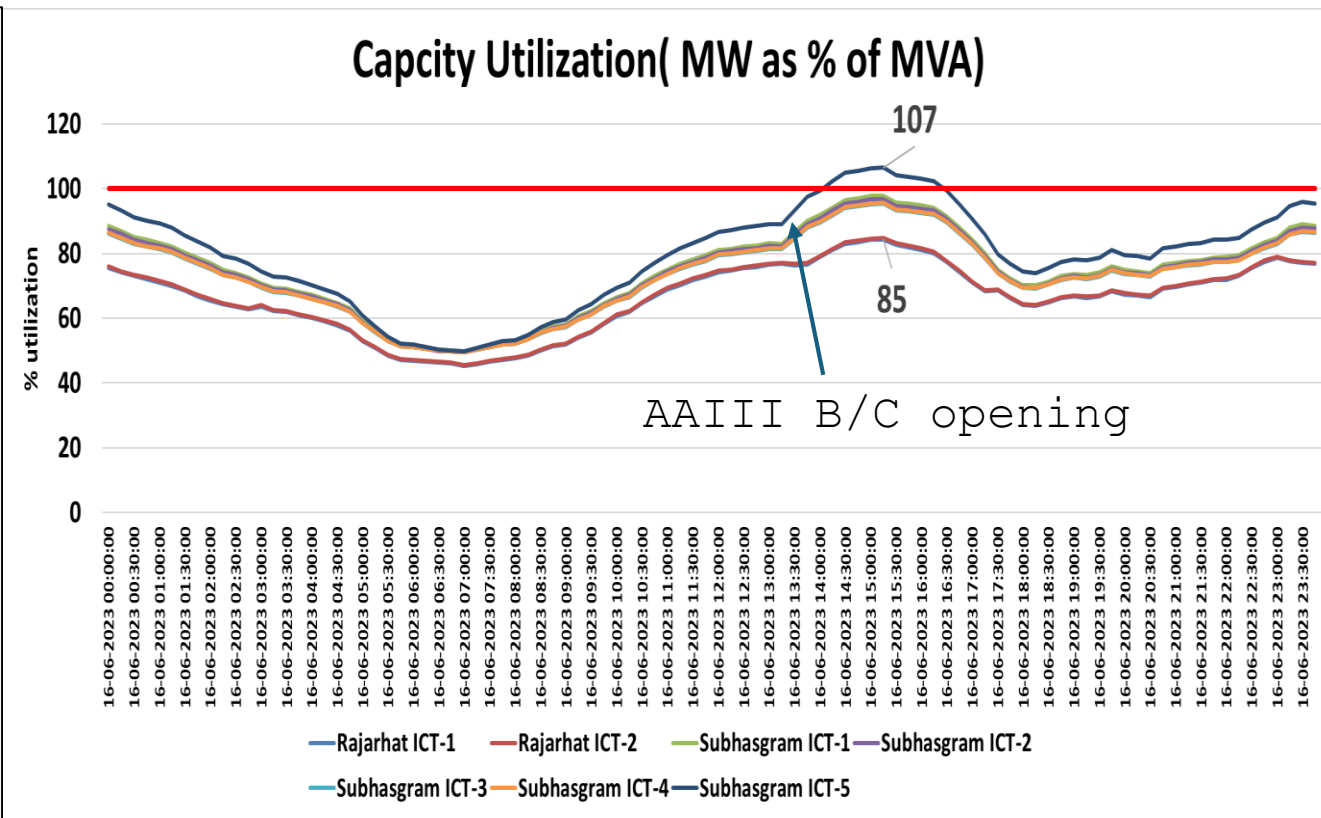
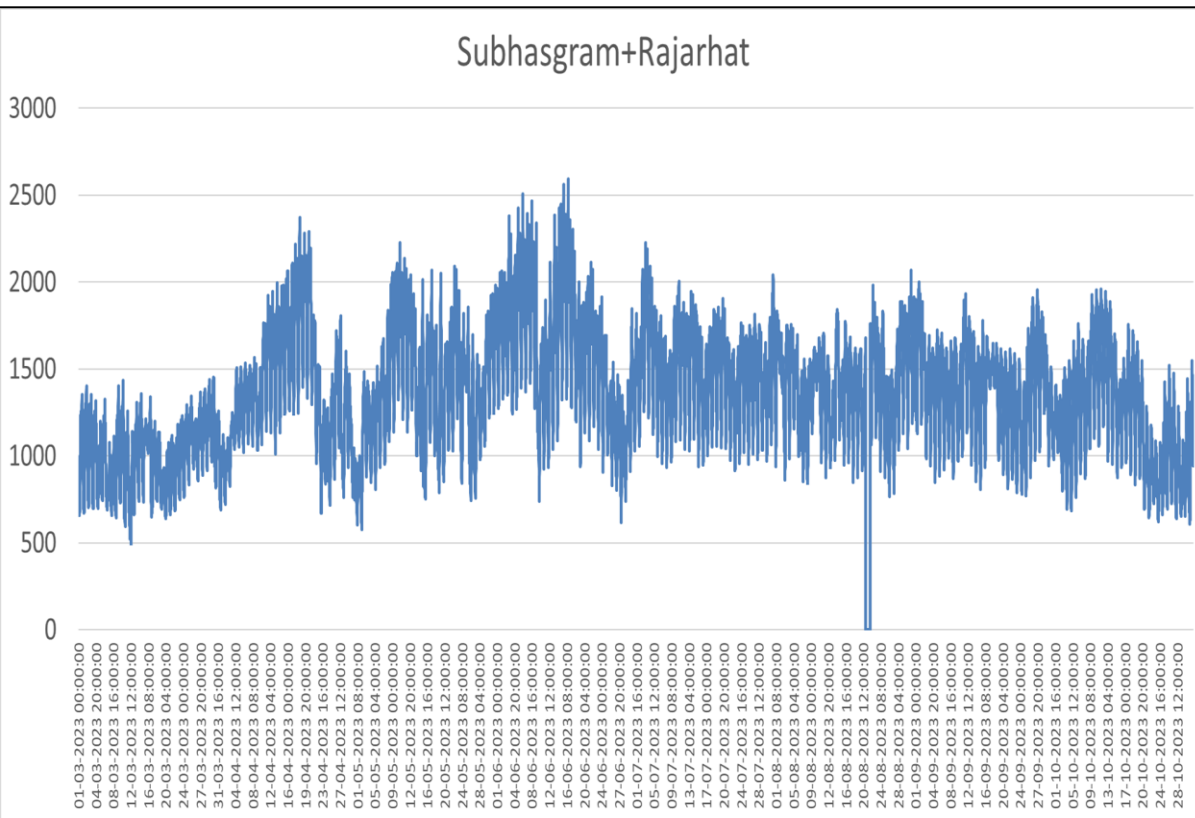
Sl. No.	Name	Designation	Organisation	Contact No.	E-mail Id	Signature
21	S. Konar	Sr. GM	ERLDC	9436335370	konar_s@grid-india.in	
22	B. Achari	DGM	ERLDC	7003472016	bilash.achari@grid-india.in	
23	Anwesh Kumar	A. Engg. C.	TVNL	8789580343	anwesh.K@tvnl.in	
24	Pinkie Debbarth	CM	ERLDC	9007079914	pinkidebbarth@grid-india.in	
25	Alok Pratek Singh		ERLDC	9007285390	apsingh@gridindia.in	
26	Chandan Mallik	Manager (SO)	ERLDC	9007059660	chandan.mallik@grid-india.in	
27	Santosh Kr. Panda	Manager	SLDC	6370134794	santosh.panda@dvce.gov.in	
28	Himadri Shekhar Sarker	Sr. Manager	DVC	8337019396	himadri.sarker@dvce.gov.in	
29	Diptikanta Panda	Manager	GMR	8114918762	Diptikanta.Panda@gmrgroup.co.in	
30	D.P. PUITANDI	Sr GM	SLDC	94341745905	Debi Prasad.puitandi@dvce.gov.in	
31	Prakanta Kumar Saha	DGM	NTPC	9437962430	Prakanta.saha@ntpc.co.in	
32	Jogeswar Majhi	GM (E)	NHPC	9800048892	jogeswar@nhpc.co.in	
33	P. V. RAUT.	AGM	MPL	9223501513	rautpv@tatapower.com	
34	Tenzin Dorji	AEE	DGPC	97517670070	t.dorji298@drnkgreen.bt	
35	Jampa Tharchen	AEE	THP, Bhutan	+97517630436	j.tharchen865@drnkgreen.bt	
36	M. S. Malik	AGM	IBEUL	6376355214	mahendra.malik@tsw.in	
37	SAIBAL GHOSH	Manager	ERLDC	8584072079	Saibal@gridindia.in	
38	Hemadri Ghally	Sr. SO	DGPC	8670690727	h.ghally2310@dgpc.bt	
39	Pigne Namgyel	AEE	DGPC	+975-17378713	j.namgyel2311@drnkgreen.bt	
40	R. Gangopadhyay	DGM	NHPC	9816681234	rathindran@nhpc.co.in	

Sl. No.	Name	Designation	Organisation	Contact No.	E-mail Id	Signature
41	D.K. Khuntia	AD	ERPC	7683889161	d.k.khunta@erpc.gov.in	
42	Saket Kumar	Sr. Mgr.	NTPC	7044474747	saketkumar@ntpc.co.in	
43	P. Ghosh	DHM	ER-D	9434748243	p.ghosh@erpc.gov.in	
44	Shiladiya Chatterjee	A.M	Adhunik Power	9748357917	shiladiya.chatterjee@adhunikpower.co.in	
45	Agniva Chatterjee	AEE	ERPC	8100307502	agniva.cea@gov.in	
46						
47						
48						
49						
50						
51						
52						
53						
54						
55						
56						
57						
58						
59						
60						

Annexure B.2.1

Overview of Last year scenario

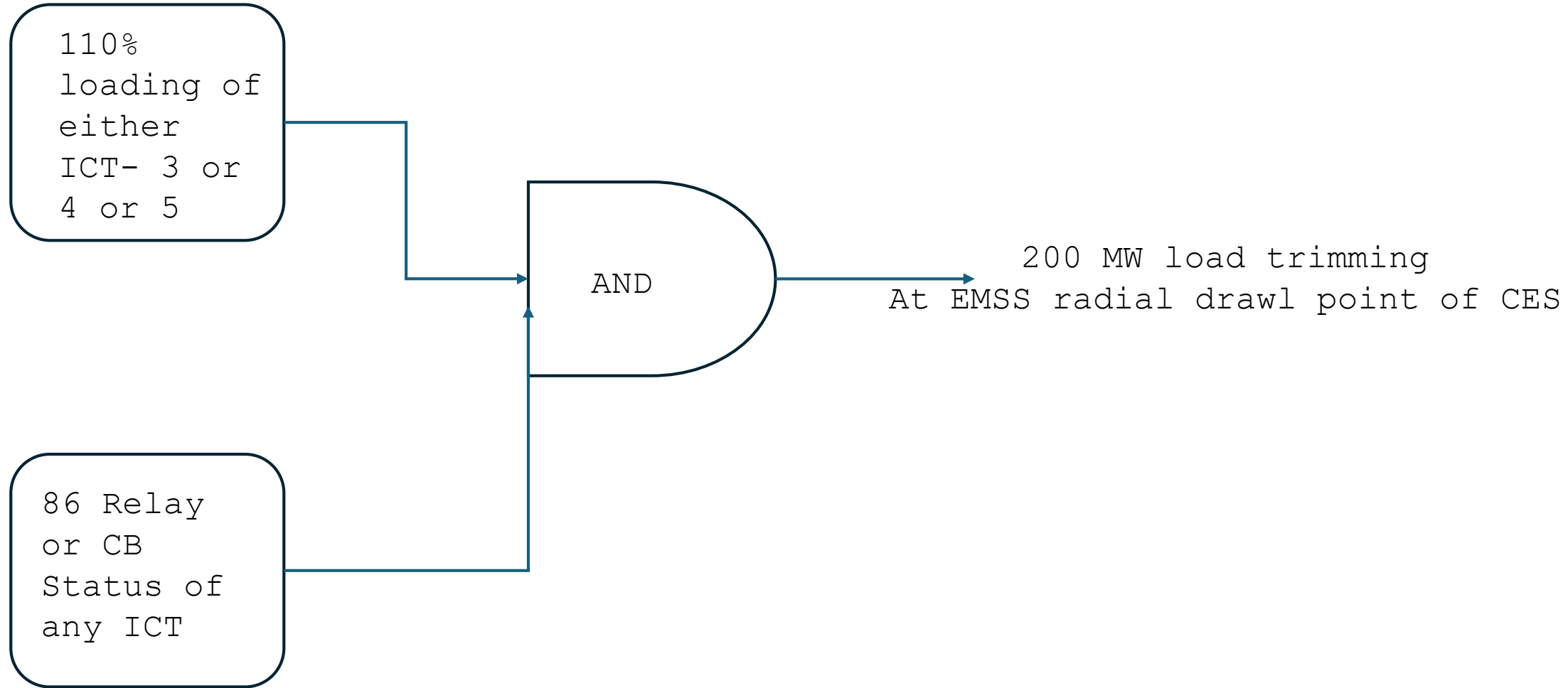
Typical hot summer day Capacity utilization last summer



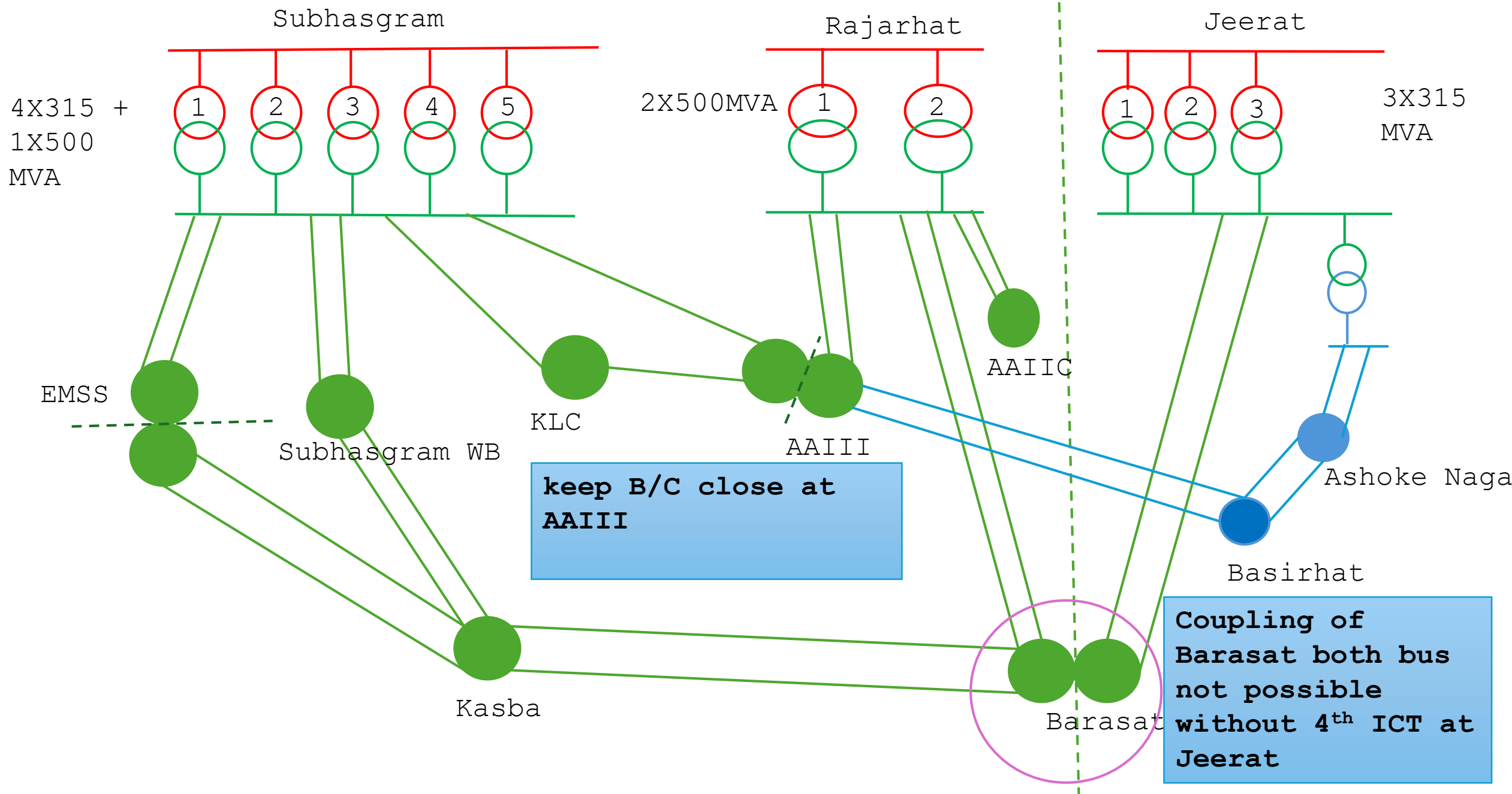
Challenges :

1. Last year Max Power flow through Subhasgram + Rajarhat ICT 2570 MW against capacity of 2760 MVA. Also, around 200 MW load restriction was there.
2. Last year due to limitation of 220 kV Rajarhat-AAIII line during peak hours, B/C at AAIII used to get opened. **This led to increase in Subhasgram ICT loading and decrease of Rajarhat ICT loading.**
3. Among the Seven ICT at Subhasgram & Rajarhat Subhasgram ICT 5 used to get beyond 100 loading , whereas little margin left in Rajarhat two ICTs

Present SPS At Subhasgram s/s



Worst Scenario: 7th ICT at Subhasgram & 4th ICT at Jeerat n



Action to be taken this year to handle worst scenario:

If we assume 8% demand growth in the area where Rajarhat & Subhasgram ICT serve loads, then we need to meet around $(2570+200)*1.08$ MW ~ 3000 MW

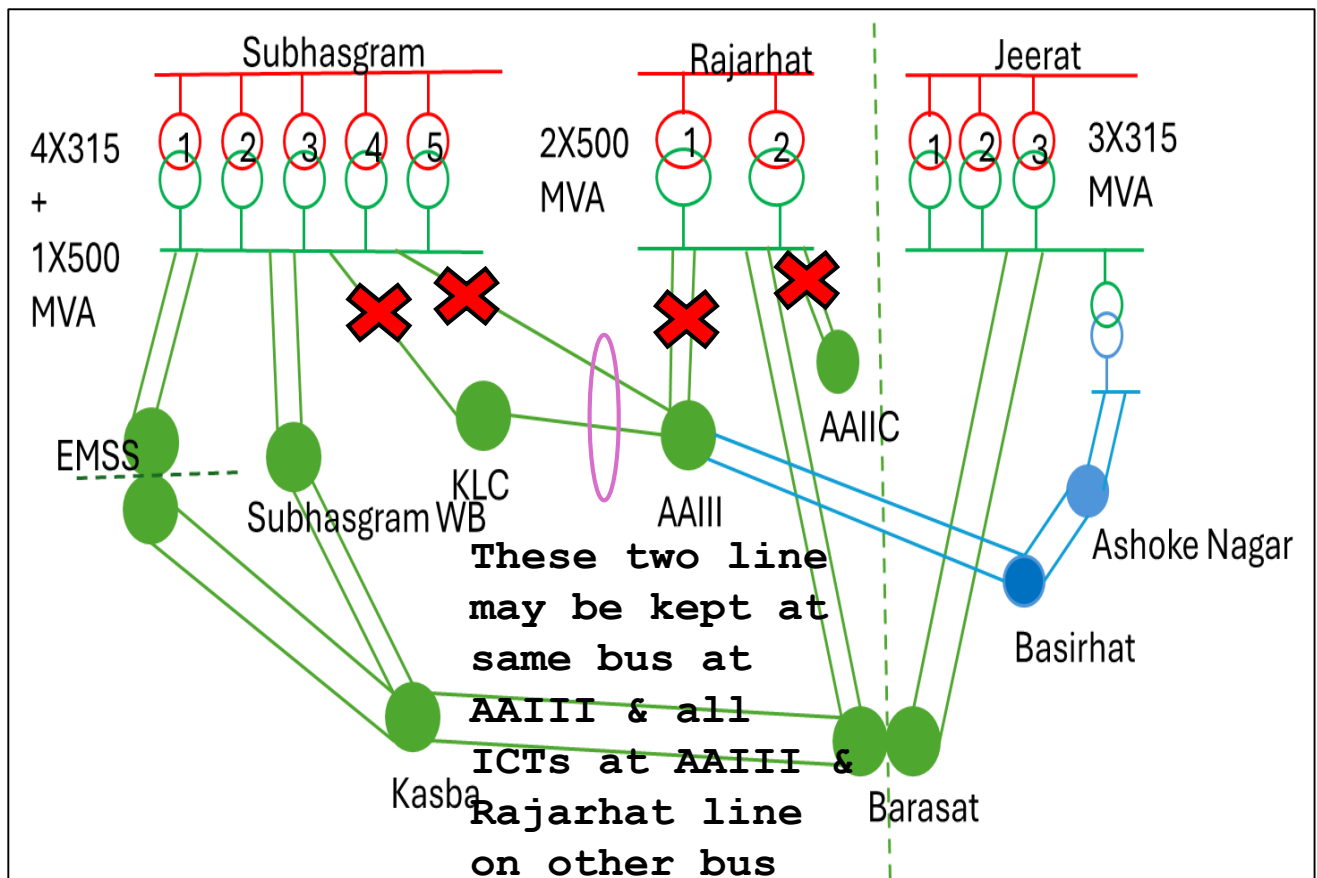
Favourable Situation:

1. As 220 kV Rajarhat-AAIII DC CT issue resolved
Therefore, AAAIII B/C can remain closed even during peak load condition
2. This will help in more uniform utilization between Rajarhat & Subhasgram ICT. Expected that with AAAIII B/C close Subhasgram ICTs utilization will reduce by 5 to 6% and Rajarhat ICT utilization will increase by 8 to 10 %
3. Still there will be some Mismatch between ICT utilization at Subhsagram itself as well as with Rajarhat.

Action point:

1. As last year already 5th ICT at Subhasgram Reached 107% loading therefore after continuous coupled bus operation AAAIII, This ICT loading will be Just within limit i.e 100 %
2. **Therefore without 7th ICT& without loading any other ICT beyond 100% we may serve Max 2600 MW. Therefore, considering projected load , restriction of 400 MW may be planned.**
3. If Rajarhat one ICT trips other ICT will be loaded 140%. Therefore, SPS is needed.
4. Also ,if Rajarhat ICT trips and its SPS did not operate then Subhasgram will run at more than 115% continuously. Still some manual shedding as Subhasgram SPS is ICT tripping linked.
5. Therefore at Subhashram SPS needs a review

Rajarhat SPS



Concern:
If Lines from Rajarhat opened but from Subhasgram did not open, Then Subhasgram ICTs may get loaded beyond 115%

ERLDC Suggestion: Along with line opening redundant tripping signal to lower kV load feeder may also be implemented. Also, at AAIH, keeping all ICT(220/132 & 33 kV) at one side, opening of B/C may be triggered by the same SPS for safety purpose.

Rajarhat SPS-1

110% loading of either ICT

86 Relay or CB Status of any ICT

This will lead to tripping of following loads

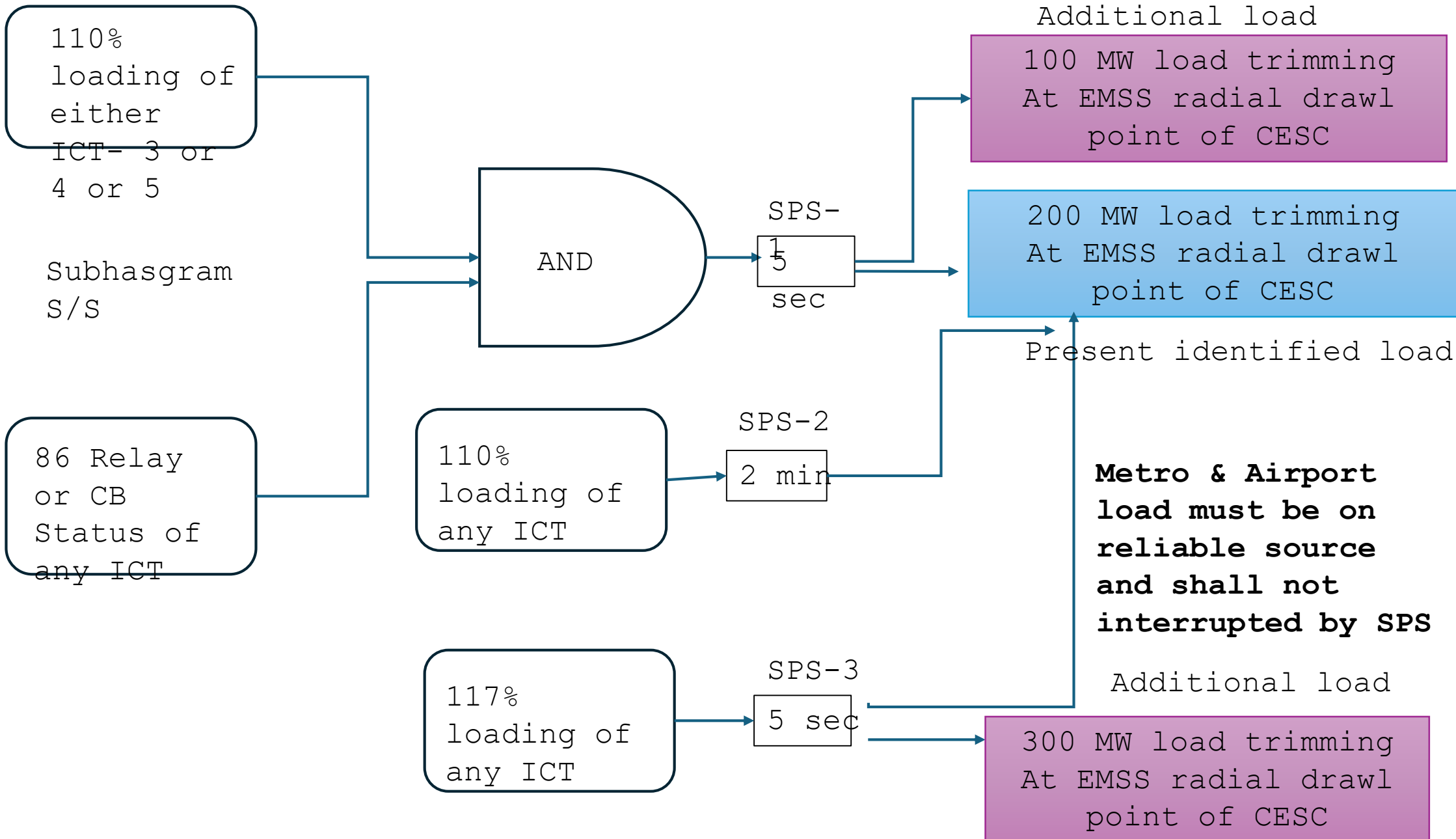
AT Rajarhat

Tripping 220 KV Rajarhat-AAIH, AAIH & SPS signal sent to Subhasgram.

At Subhasgram
Tripping of KLC line & AAIH line

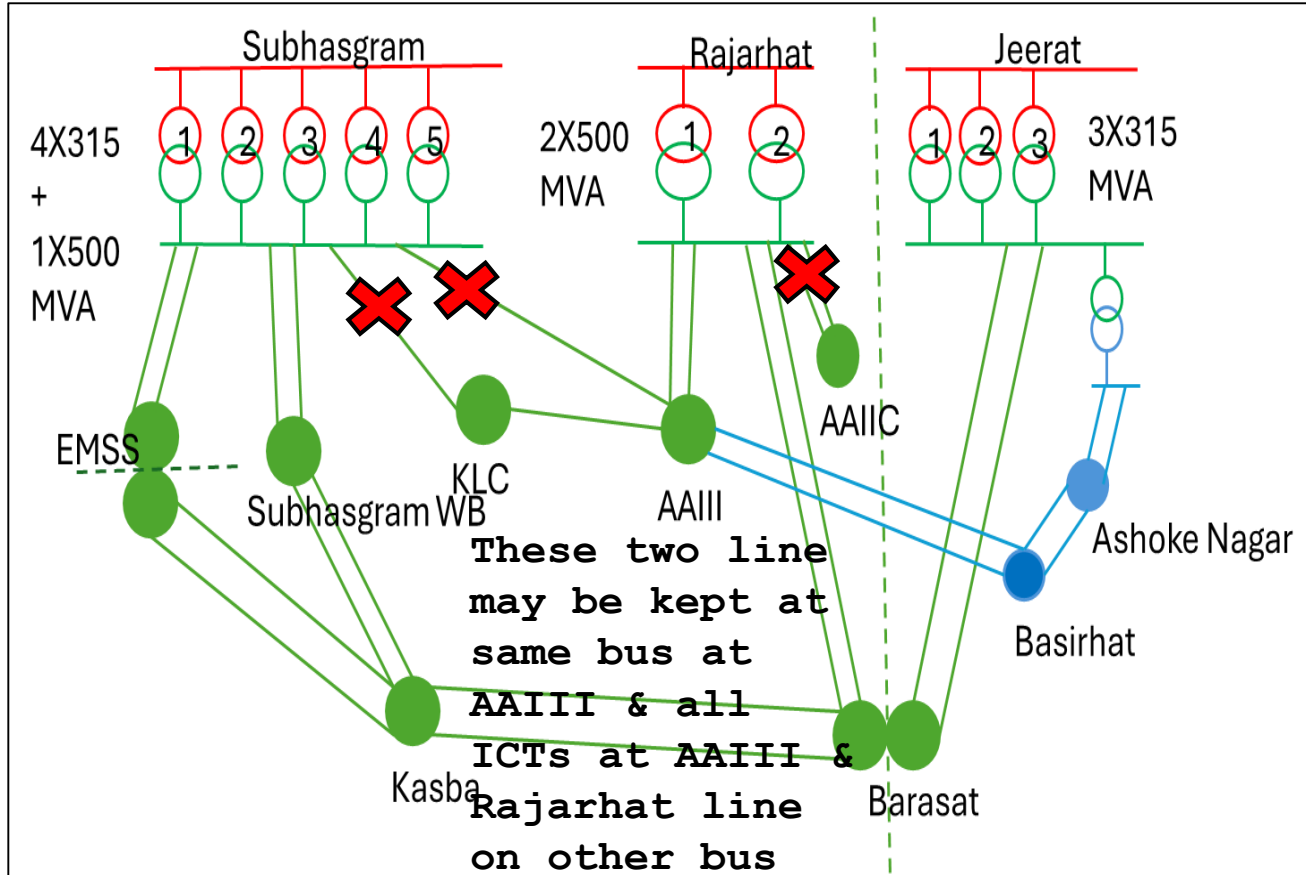
Substation	Load
CLC	100
Salt lake GIS	70
Action area I	79
Action area II	60
Action area III	124

Proposed SPS at SUBHASGRAM



With Barasat
Bus coupled

Rajarhat SPS



Rajarhat SPS-1

110%
loading of
either ICT

At Rajarhat
Tripping
220 KV
Rajarhat-
AAII, .

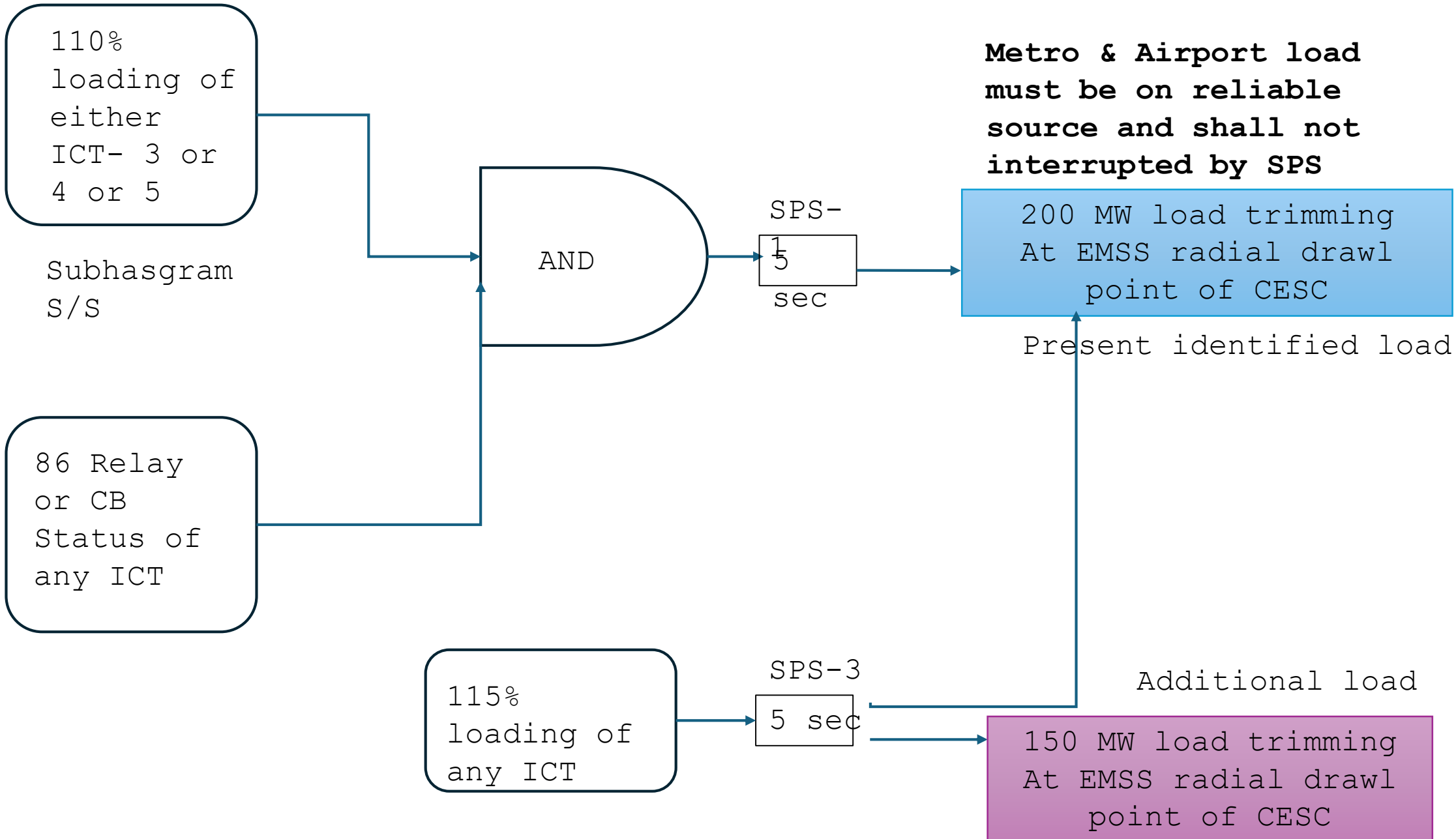
AND

At Subhasgram
Tripping of
KLC line &
AAIII line

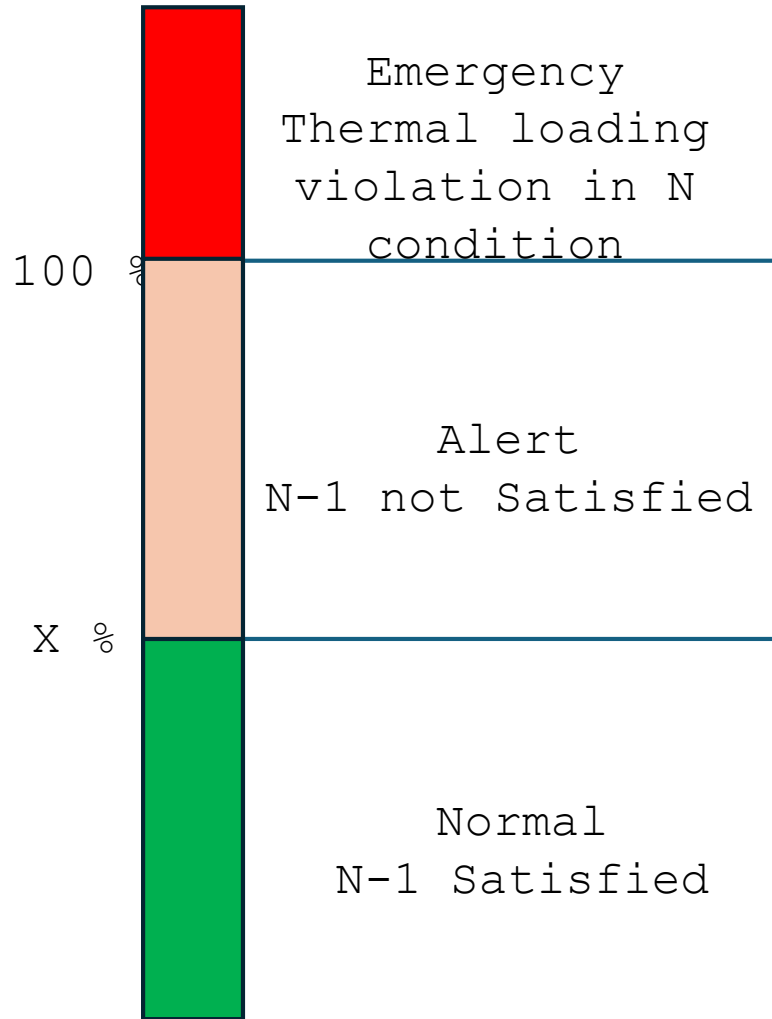
86 Relay
or CB
Status of
any ICT

At AAIII
Tripping of
220/33 kV
load

Proposed SPS at SUBHASGRAM

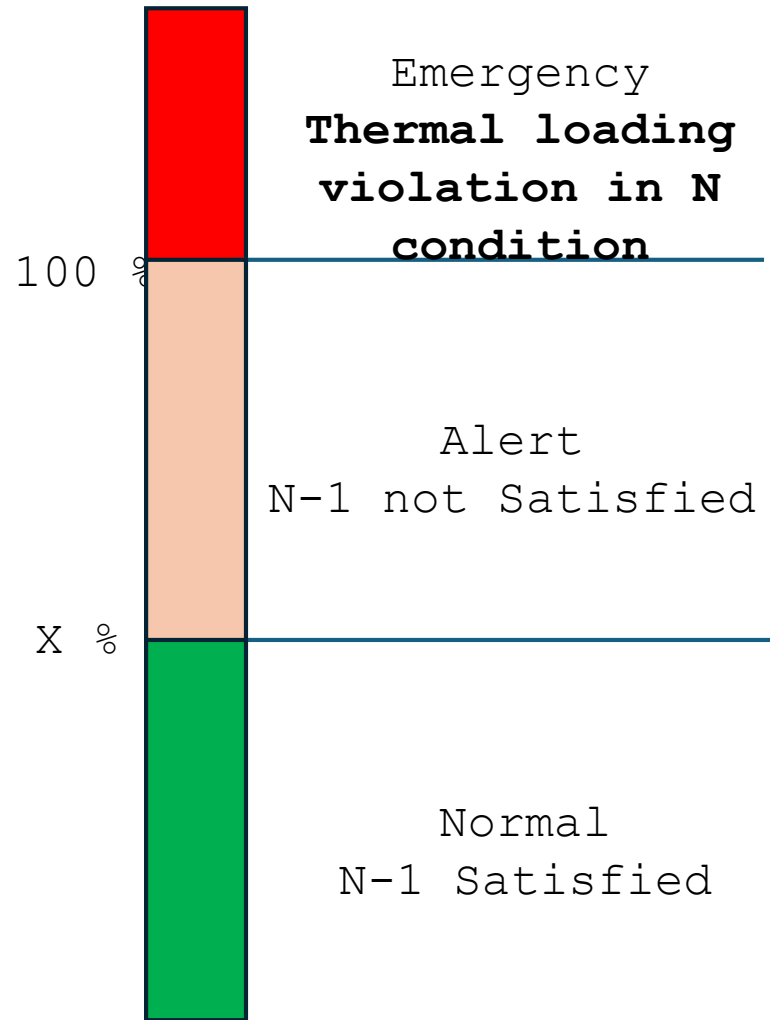


Basis for SOP Design



Point under consideration:

1. Failure of SPS or insufficient action
2. Emergency as per operating procedure without any tripping.



Condition: Overloading of Subhasgram ICT beyond 105%:

Action by ERLDC:

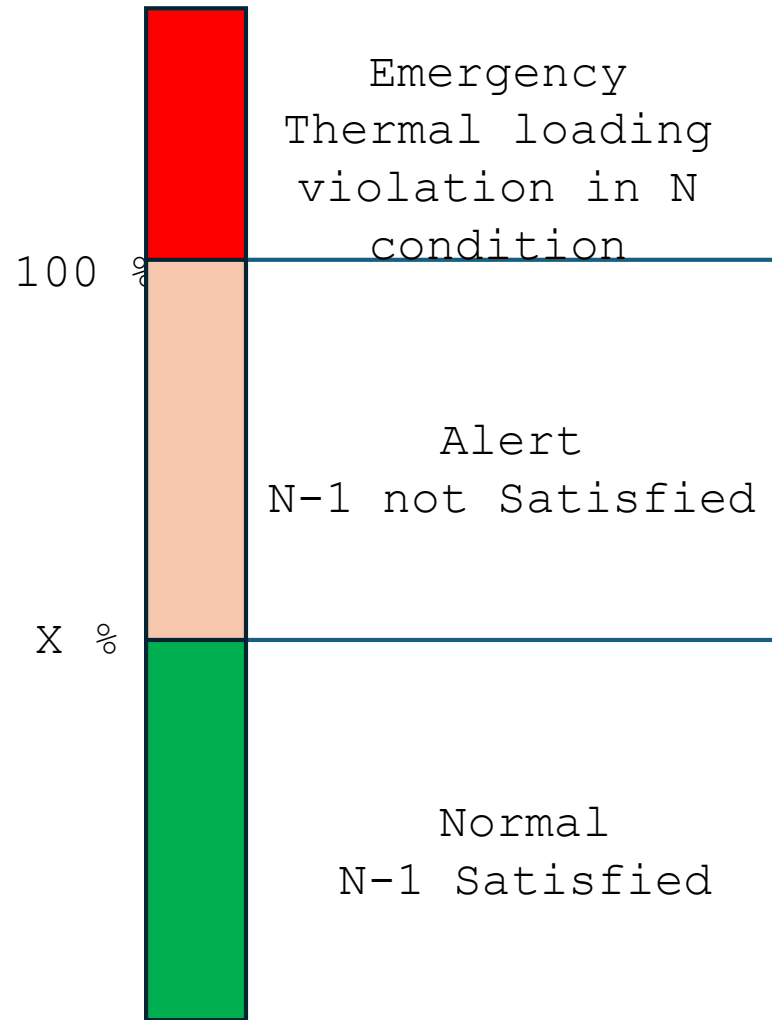
As any element loading beyond 100% means an emergency situation, therefore load trimming instruction will be issued to WBSLDC.

Action by WBSLDC:

Immediate action to control drawl within 100% in coordination with CESC.

Action by CESC:

Identify non-critical load which are not included in any SPS scheme and do manual load trimming



Condition: Overloading of Subhasgram ICT beyond 117% due to failure of SPS operation with any ICT tripping at Subhasgram:

Action by ERLDC:

Issue immediate load trimming instruction to WBSLDC & CESC. In case loading did not come below 110% within 5 min, Opening code will be issued to POWERGRID for 220 kV Subhsagram-Baruipur followed by 220 kV Subhasgram-EMSS line opening if load did not come within 110%.

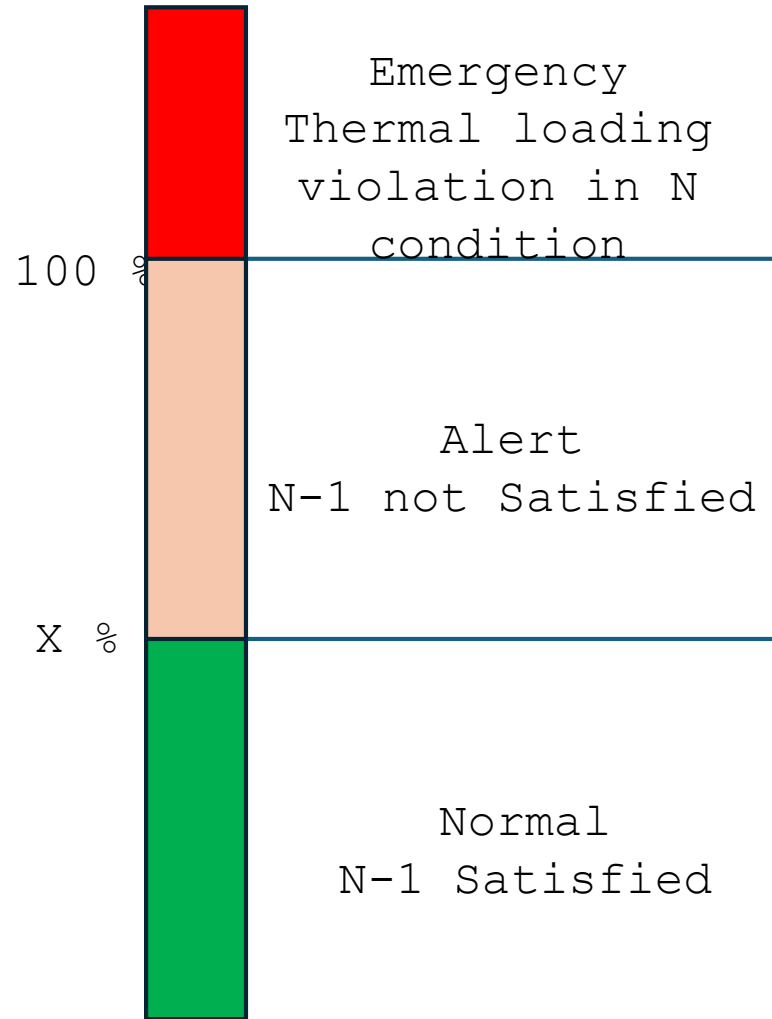
Action by WBSLDC:

Immediate action to control drawl within 100% in coordination with CESC.

Action by CESC:

Manually tripping of identified load for SPS.

SOP-2 considering Barasat Bus coupled



Condition: Overloading of Subhasgram ICT beyond 117% due to failure of SPS operation with any ICT tripping at Subhasgram:

Action by ERLDC:

Issue immediate load trimming instruction to WBSLDC & CESC. In case loading did not come below 110% within 5 min, Line opening decision may be taken. WBSLDC & CESC may confirm the line.

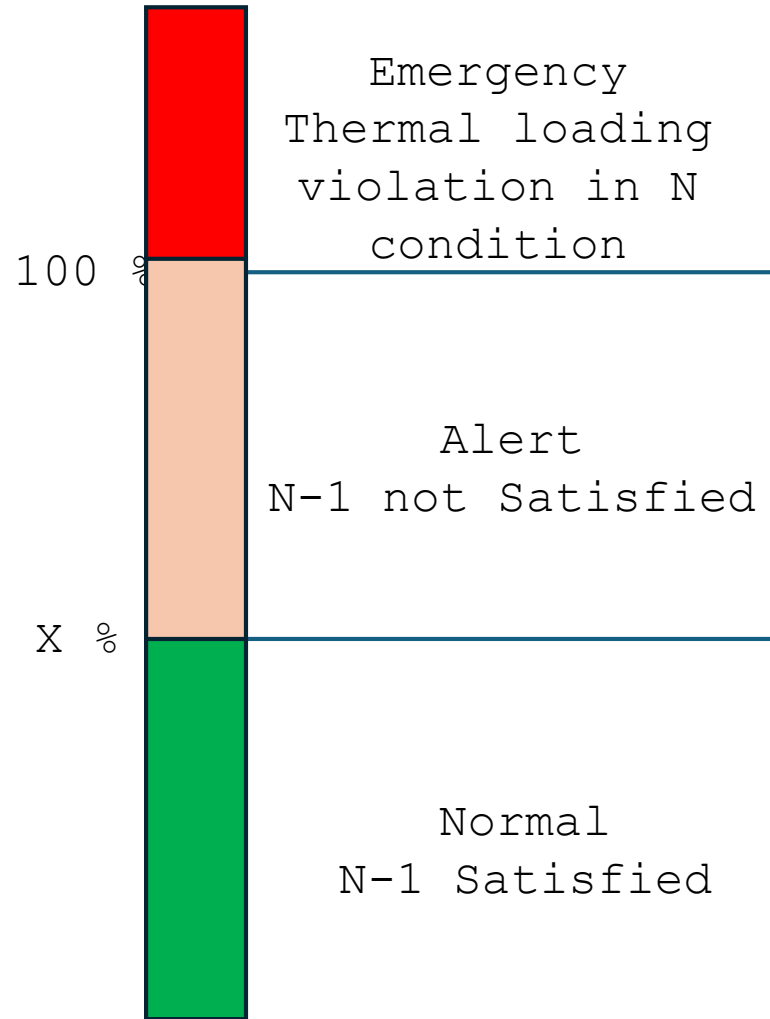
Action by WBSLDC:

Immediate action to control drawl within 100% in coordination with CESC.

Action by CESC:

Manually tripping of identified load for SPS.

SOP-3



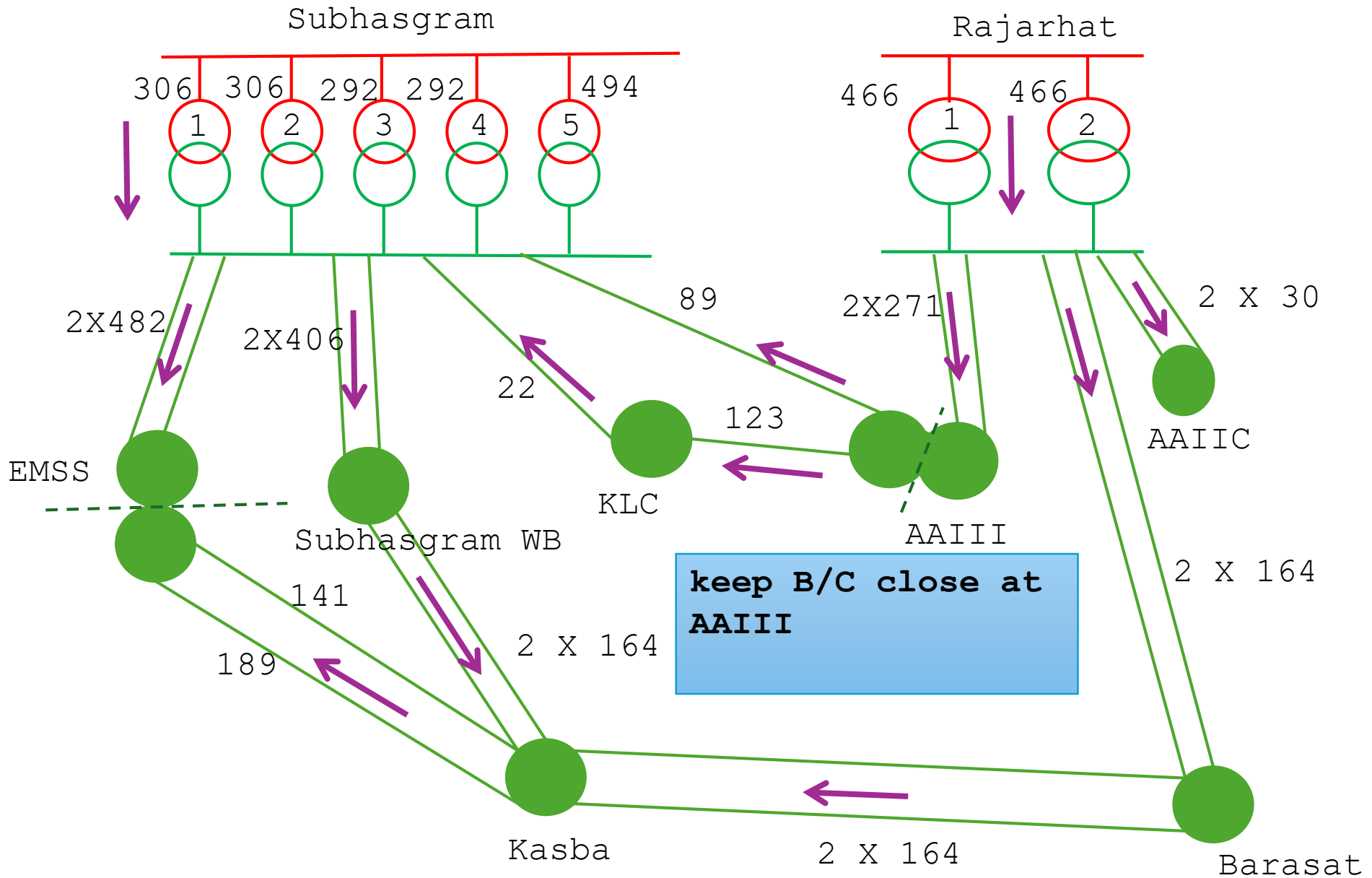
Condition: Overloading of Subhasgram ICT beyond 117% due to failure of SPS operation with 500 MVA ICT tripping at Rajarhat:

Action by ERLDC , WBSLDC & POWERGRID:
Immediate mimic the SPS logic manually.

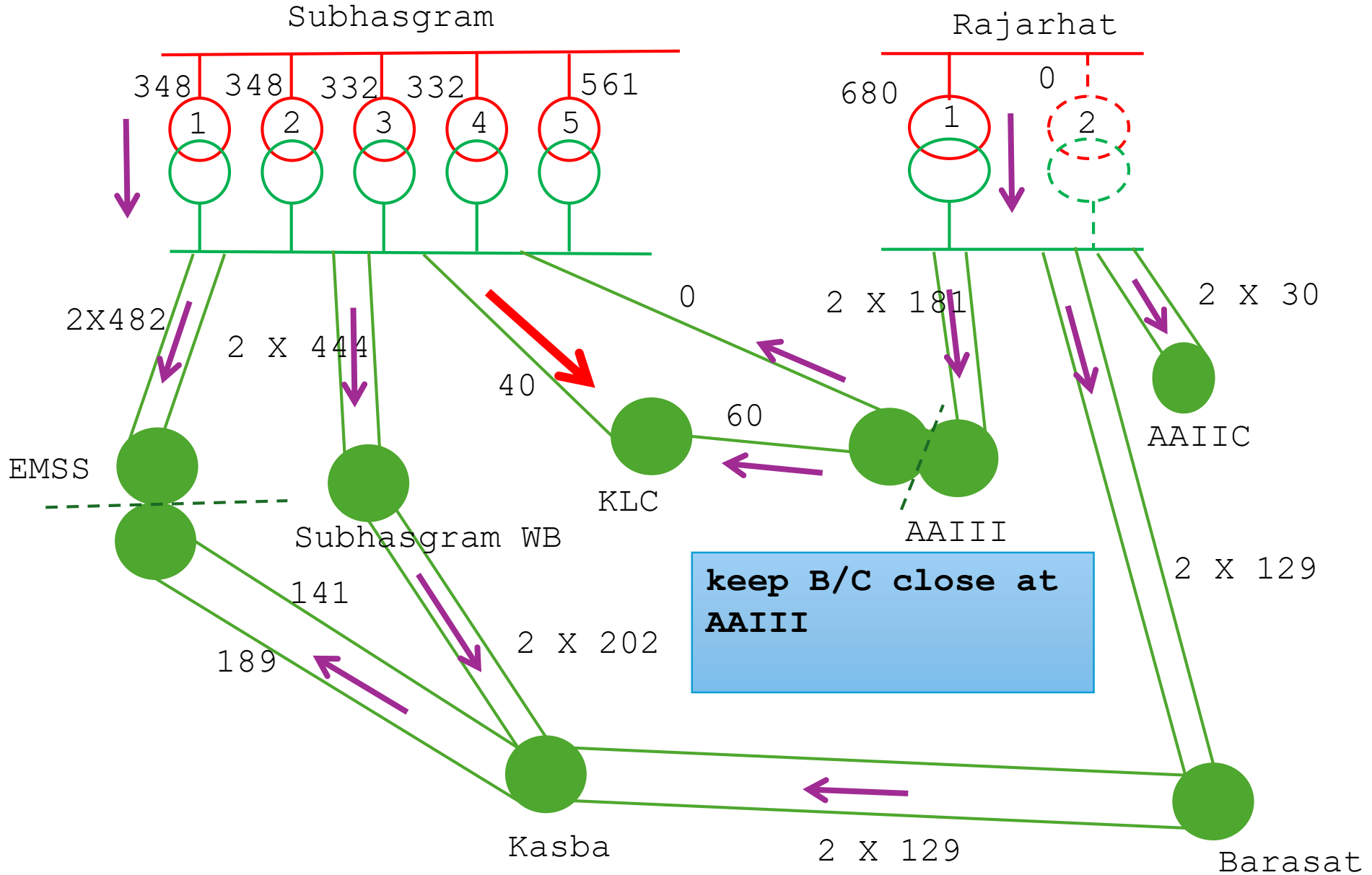
Annexures

Power Flows

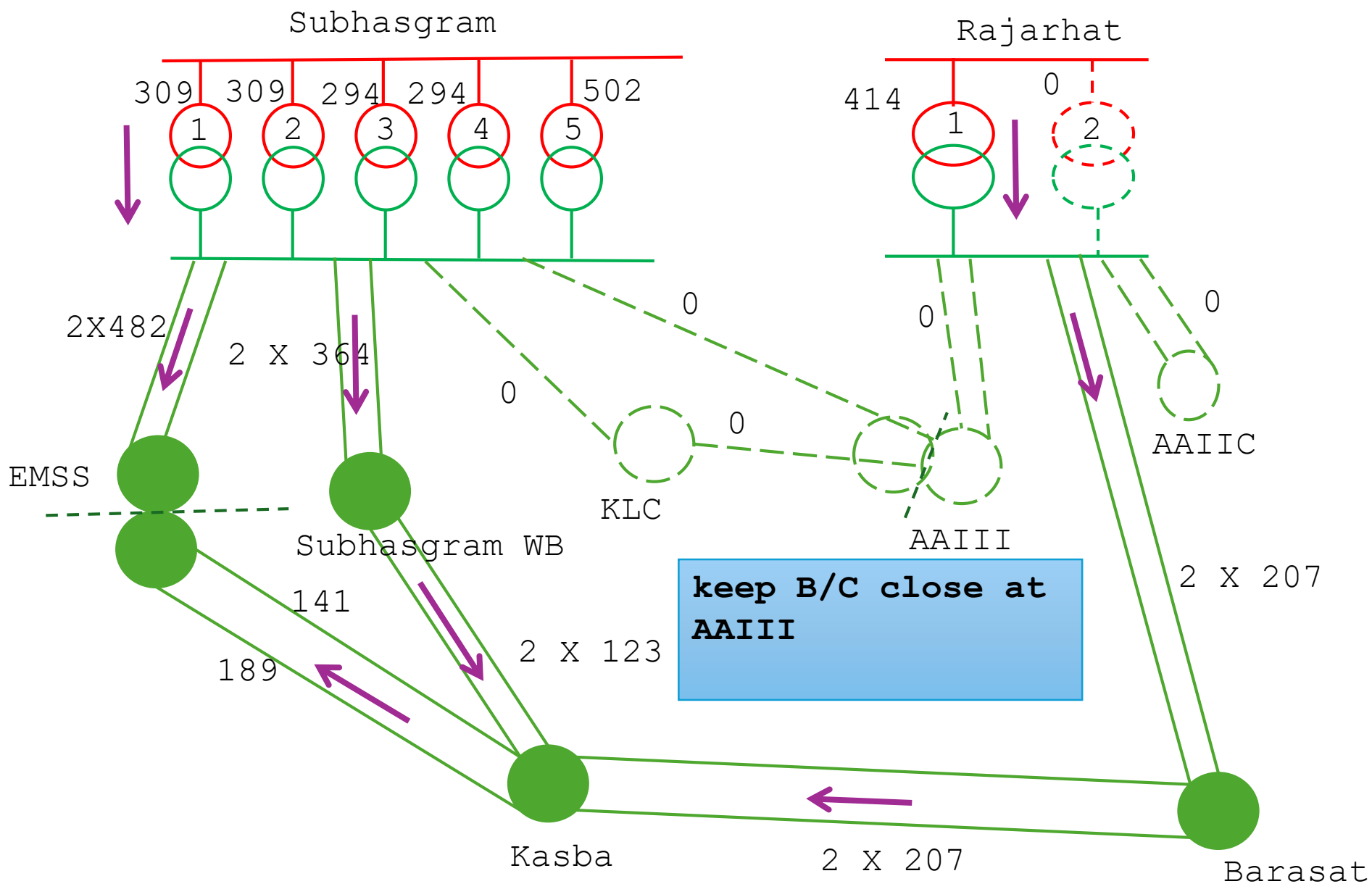
Base Case flows



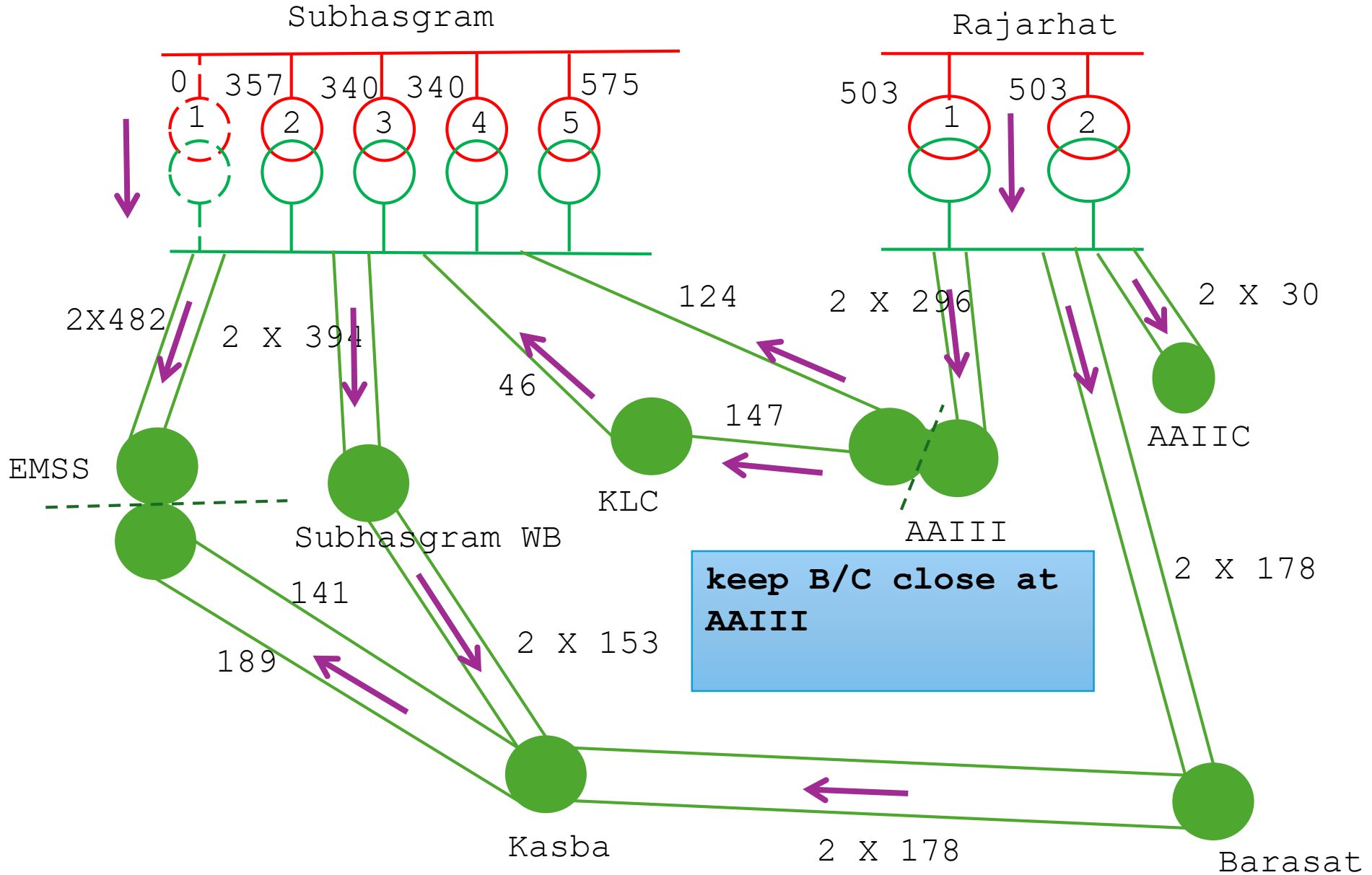
Flow after Rajarhat ICT



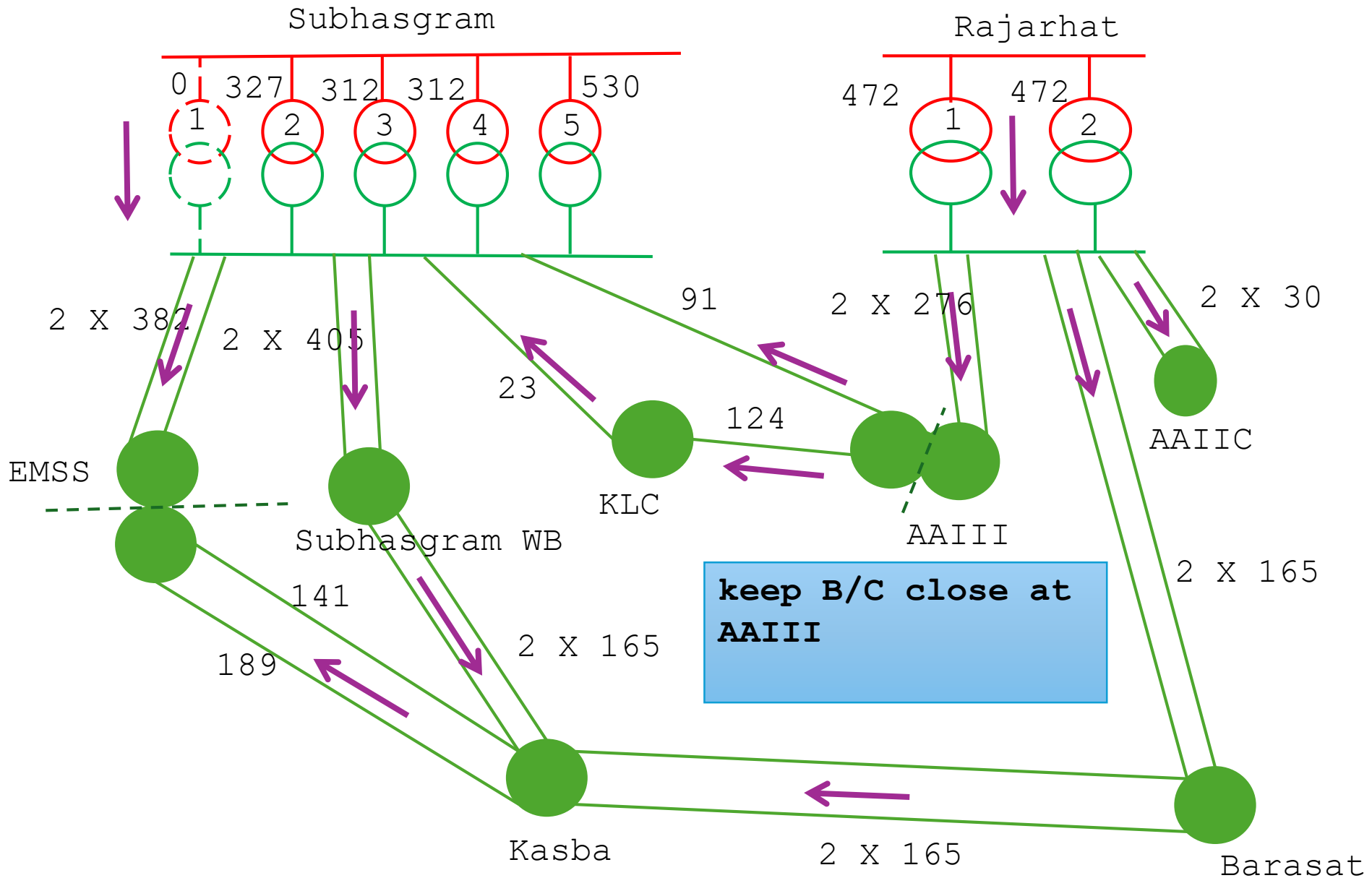
Flow after Rajarhat ICT Tripping & Rajarhat



Subhasgram 315 MVA ICT Tri

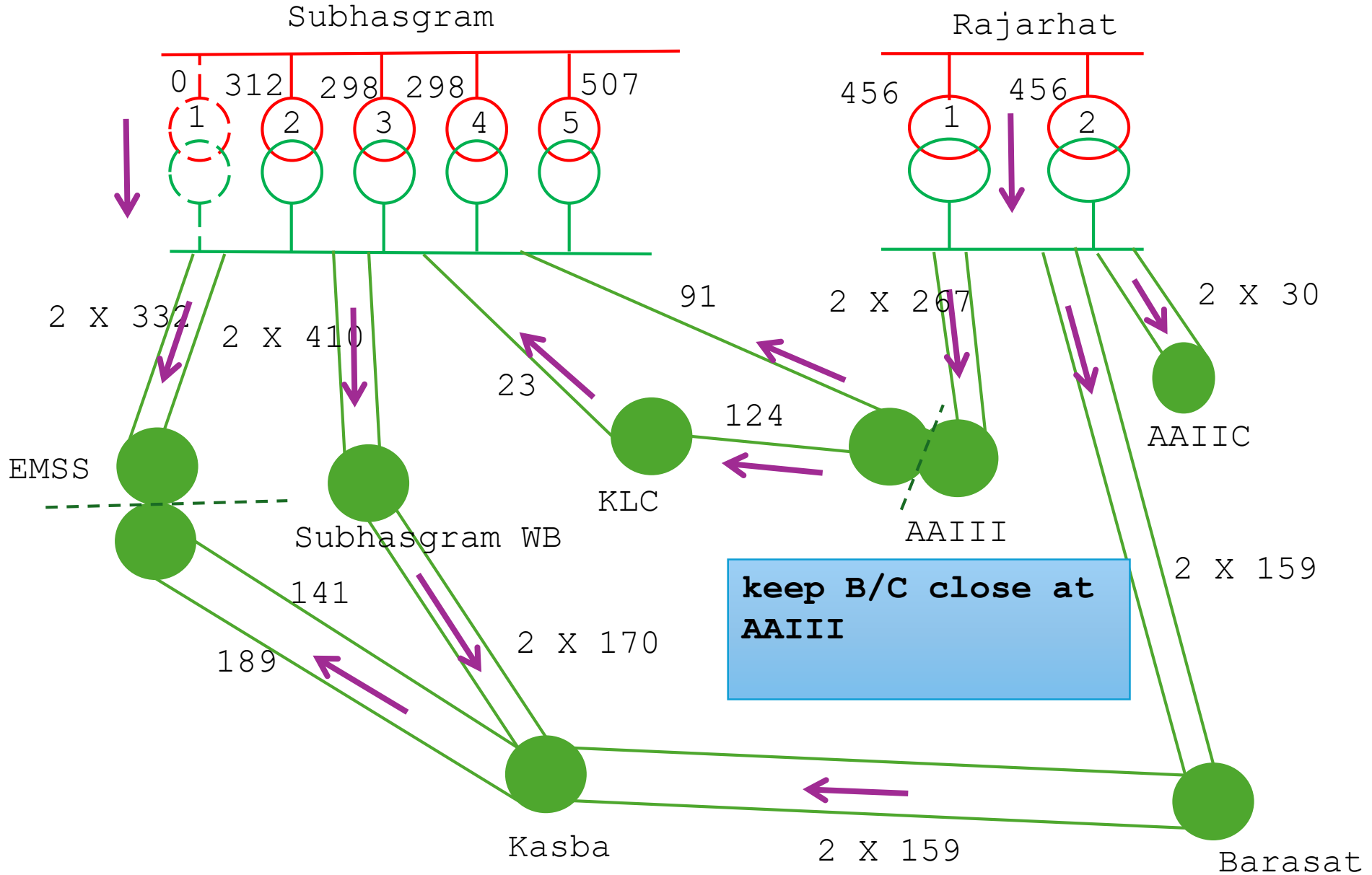


Subhasgram 315 MVA ICT Tripping & Present SPS operation at Subhasgram



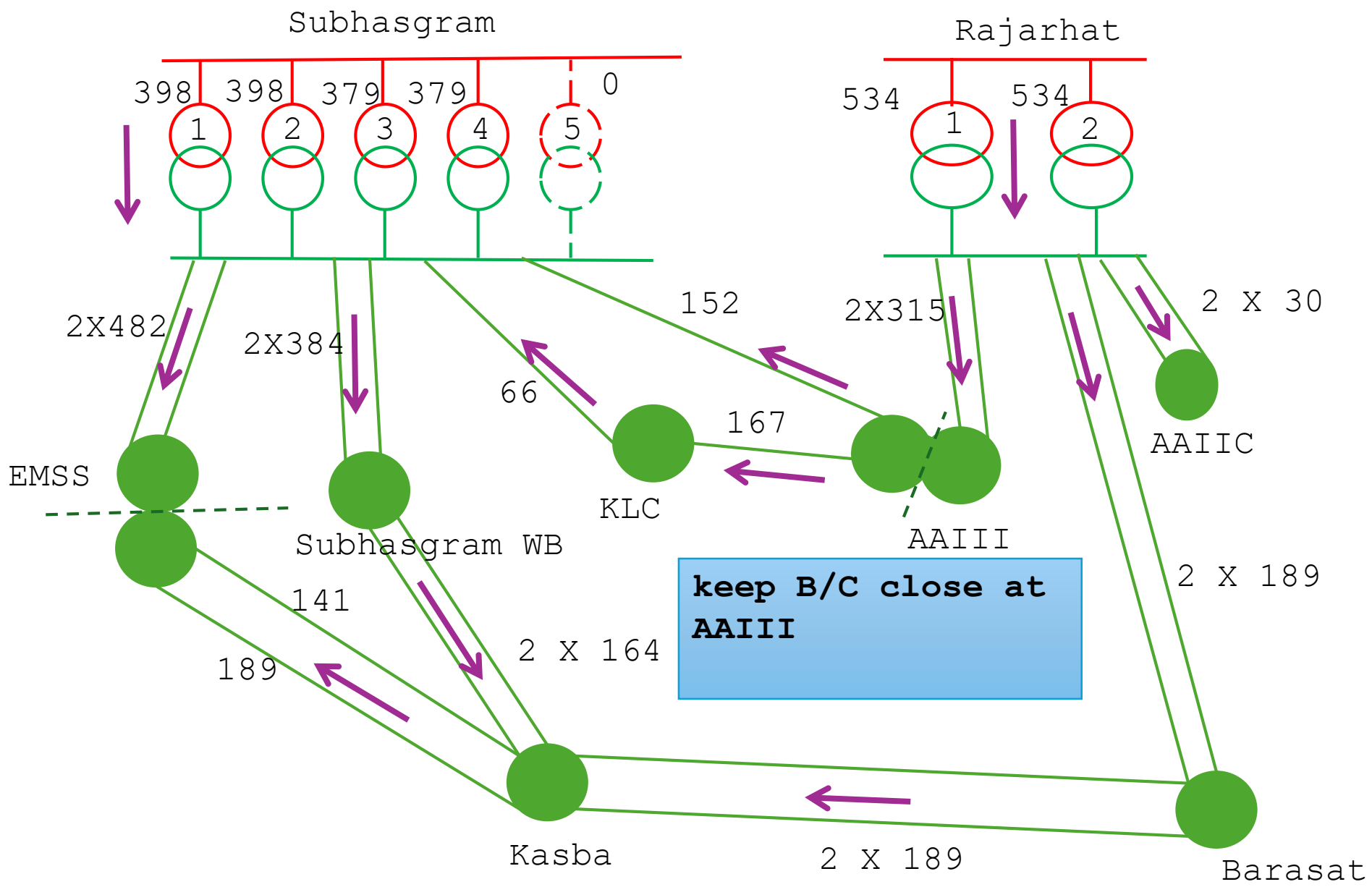
Insufficient to control Subhasgram ICT loading

Subhasgram 315 MVA ICT Tripping & Proposed SPS operation at

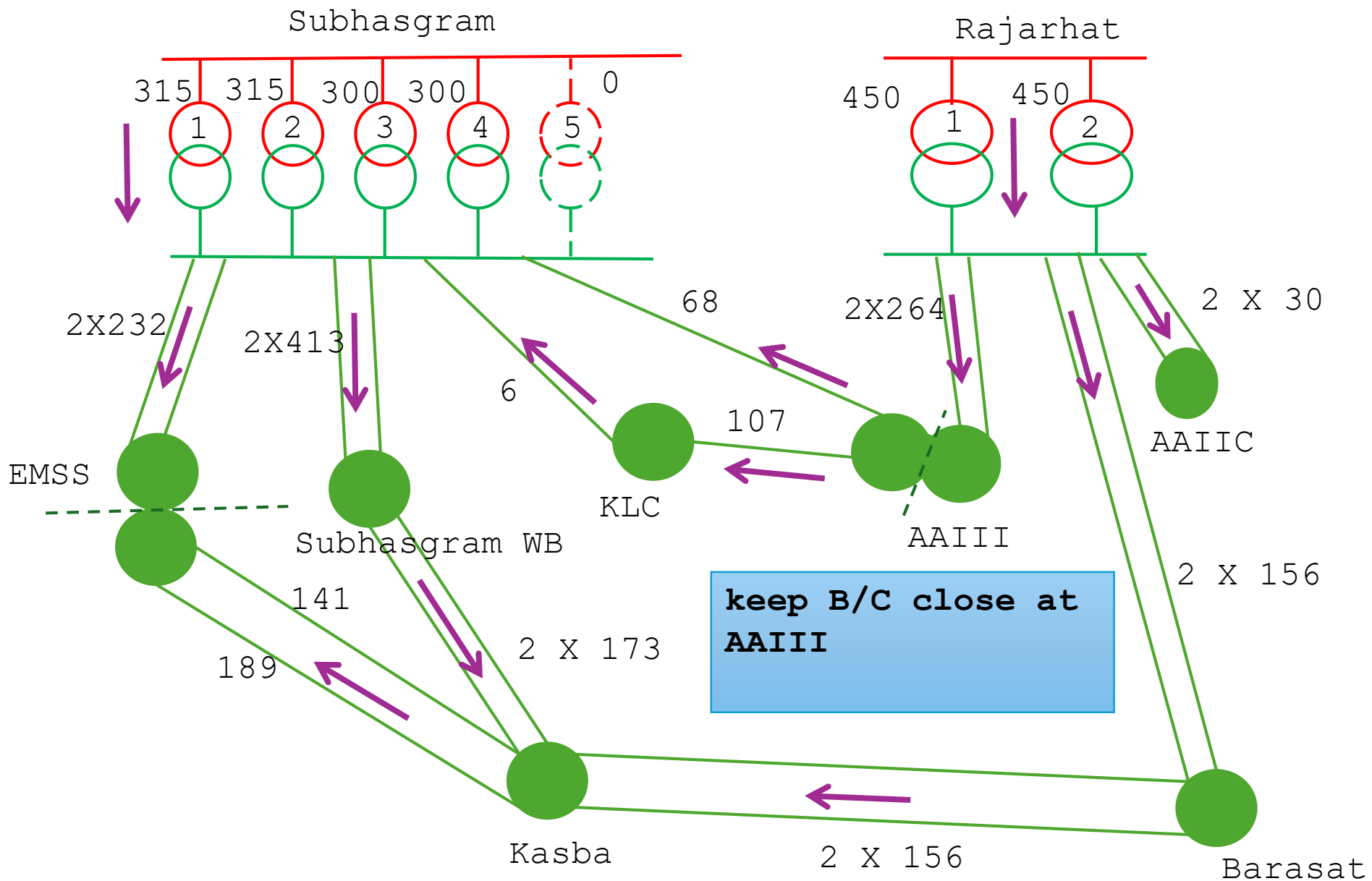


Subhasgram ICT loading within limit

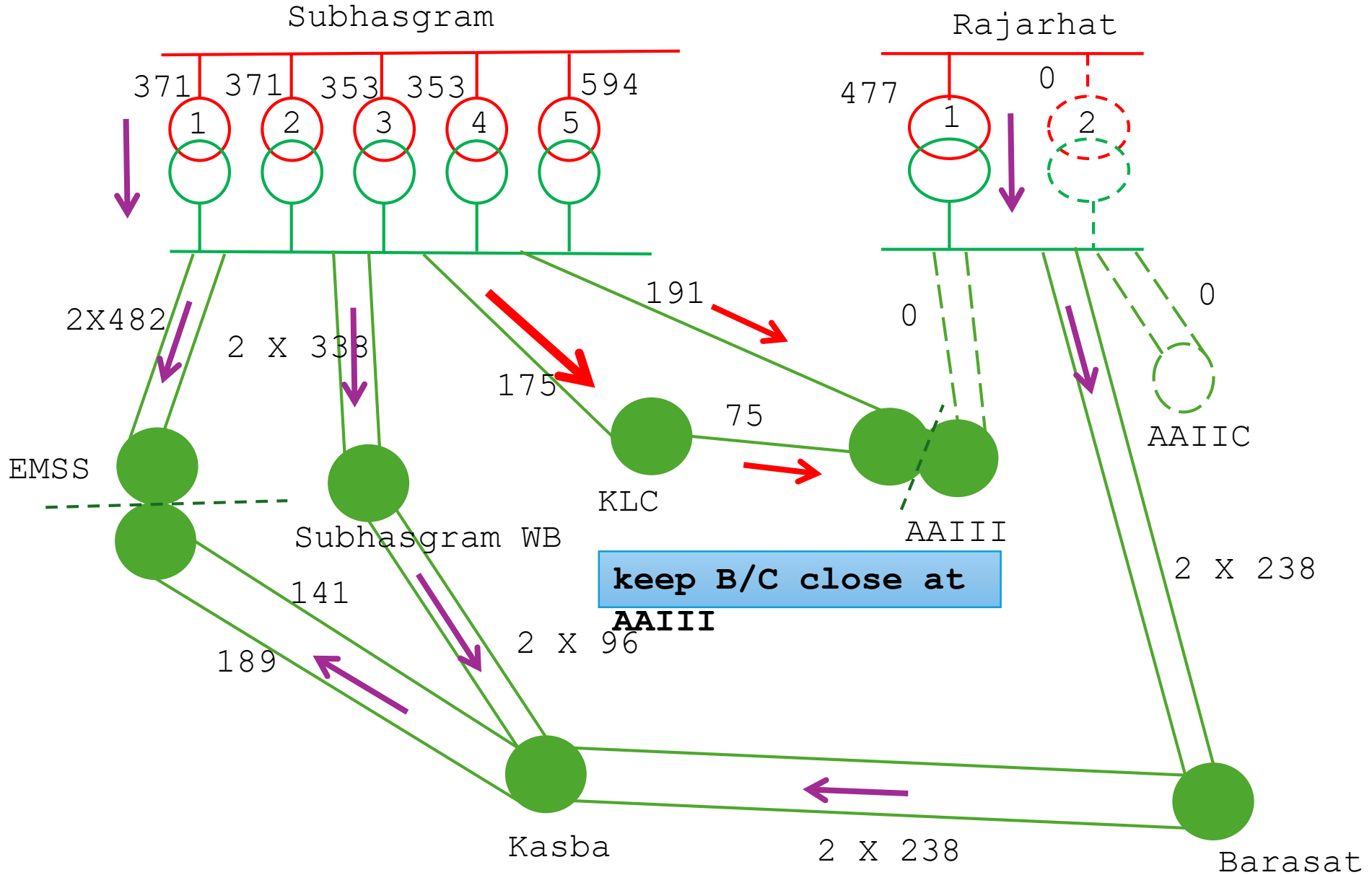
Subhasgram 500 MVA ICT Tri



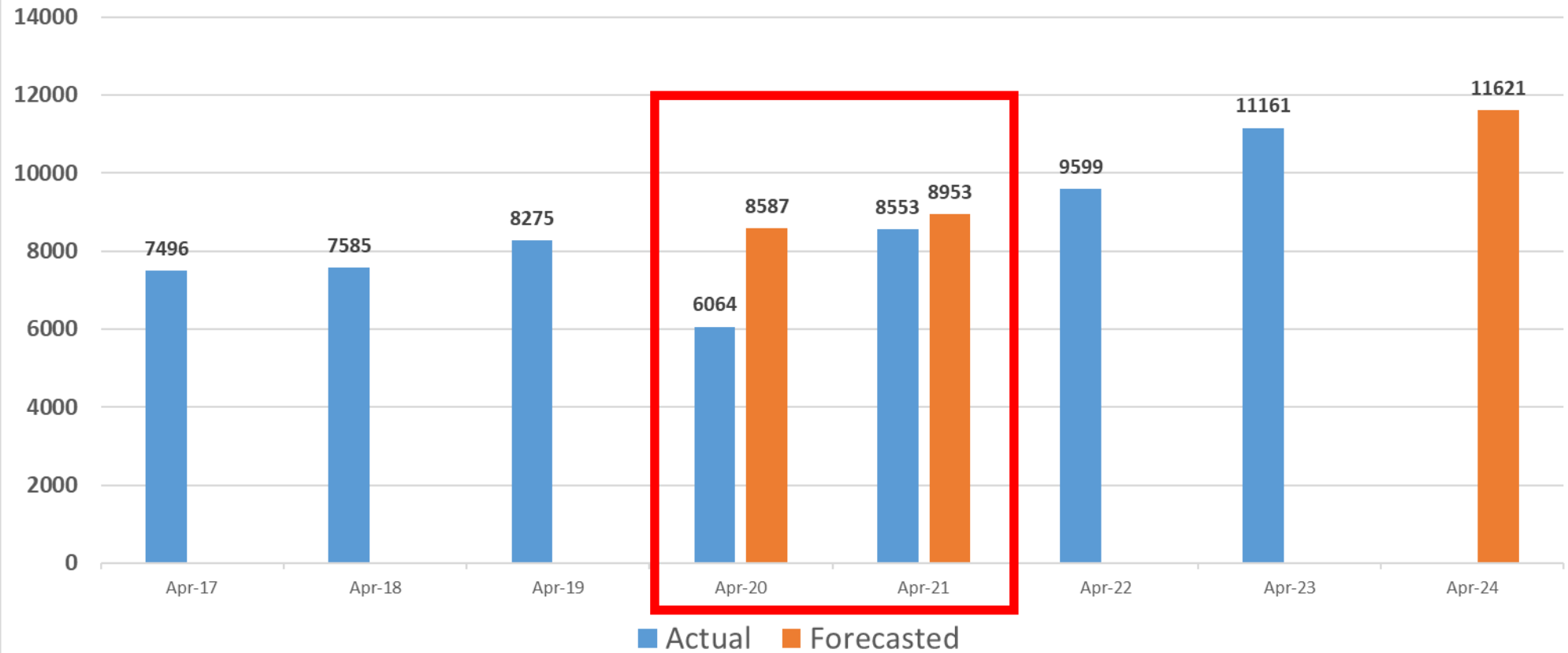
Subhasgram 500 MVA ICT Tripping & Proposed SPS operation at



Flow after Rajarhat ICT Tripping & Opening of lines from



West Bengal Max Demand During April at 15:00 Hrs



Annexure D.1

Updated Anticipated Peak Demand (in MW) of ER & its constituents for April 2024

1	BIHAR	Demand (MW)	Energy Requirement (MU)
	NET MAX DEMAND	6602	3625
	NET POWER AVAILABILITY- Own Sources	497	353
	Central Sector+Bi-Lateral	6567	4112
	SURPLUS(+)/DEFICIT(-)	462	840
2	JHARKHAND		
	NET MAXIMUM DEMAND	2100	1134
	NET POWER AVAILABILITY- Own Source	350	200
	Central Sector+Bi-Lateral+HPP	1158	721
	SURPLUS(+)/DEFICIT(-)	-592	-213
3	DVC		
	NET MAXIMUM DEMAND	3300	2160
	NET POWER AVAILABILITY- Own Source	5500	3664
	Central Sector+MPL	300	164
	Bi- lateral export by DVC	2500	1680
	SURPLUS(+)/DEFICIT(-) AFTER EXPORT	0	-12
4	ODISHA		
	NET MAXIMUM DEMAND (OWN)	5200	3240
	NET MAXIMUM DEMAND (In Case of CPP Drawal)	6100	3888
	NET POWER AVAILABILITY- Own Source	3270	1951
	Central Sector	1930	1289
	SURPLUS(+)/DEFICIT(-) (OWN)	0	0
	SURPLUS(+)/DEFICIT(-) (In Case, 900 MW CPP Drawal)	-900	-648
5	WEST BENGAL		
	WBSEDCL		
5.1	NET MAXIMUM DEMAND	9500	4998
	NET MAXIMUM DEMAND (Incl. Sikkim)	9505	5002
	NET POWER AVAILABILITY- Own Source (Incl. DPL)	5425	3126
	Central Sector+Bi-lateral+IPP&CPP+TLDP	2523	1234
	EXPORT (To SIKKIM)	5	4
	SURPLUS(+)/DEFICIT(-) AFTER EXPORT	-1557	-642
5.2	CESC		
	NET MAXIMUM DEMAND	2565	1142
	NET POWER AVAILABILITY- Own Source	830	549
	IMPORT FROM HEL	541	376
	TOTAL AVAILABILITY OF CESC	1371	925
	DEFICIT(-) for Import	-1194	-217
	WEST BENGAL (WBSEDCL+CESC+IPCL)		
	(excluding DVC's supply to WBSEDCL's command area)		
	NET MAXIMUM DEMAND	12351	6140
	NET POWER AVAILABILITY- Own Source	6255	3675
	CS SHARE+BILATERAL+IPP/CPP+TLDP+HEL	3064	1610
	SURPLUS(+)/DEFICIT(-) BEFORE WBSEDCL'S EXPORT	-2746	-855
	SURPLUS(+)/DEFICIT(-) AFTER WBSEDCL'S EXPORT	-2751	-859
6	SIKKIM		
	NET MAXIMUM DEMAND	118	52
	NET POWER AVAILABILITY- Own Source	49.6	60
	Central Sector	114	70
	SURPLUS(+)/DEFICIT(-)	46	78
	EASTERN REGION		
	NET MAXIMUM DEMAND	29385	16351
	NET MAXIMUM DEMAND (In Case of CPP Drawal of Odisha)	30285	16999
	BILATERAL EXPORT BY DVC (Incl. Bangladesh)	2333	1680
	EXPORT BY WBSEDCL TO SIKKIM	5	4
	EXPORT TO B'DESH & NEPAL OTHER THAN DVC	642	462
	NET TOTAL POWER AVAILABILITY OF ER	27971	17005
	(INCLUDING CS ALLOCATION +BILATERAL+IPP/CPP+HEL)		
	SURPLUS(+)/DEFICIT(-)	-4394	-1492
	SURPLUS(+)/DEFICIT(-) (In Case of CPP Drawal for Odisha)	-5294	-2140